

Cell Structure and Function

Chapter-03:
Cell Structure and Function
Course BIO-3105: Biology for
Engineers
Lecture-4+5+6;
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Ref book: Biology for Engineers - Arthur T. Johnson [2nd edition]
Biology for Engineers – G. K. Suraishkumar

Cell Structure and Function : Overview

- Prokaryotic and Eukaryotic cells
- Plant and animal cells
- Cell Structure and Function
- Fluid mosaic model
- Protein factories of the cell
- Nucleated and enucleated cells

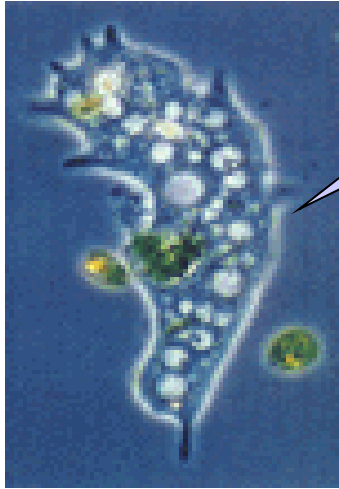
Cell Theory:

- All living things are made up of cells
- Cells are the smallest working units of all living things
- All cells come from preexisting cells through cell division

Definition of Cell:

A cell is the smallest unit that is capable of performing life functions.

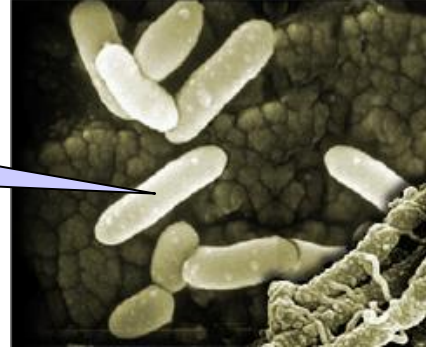
Some Examples of Cells



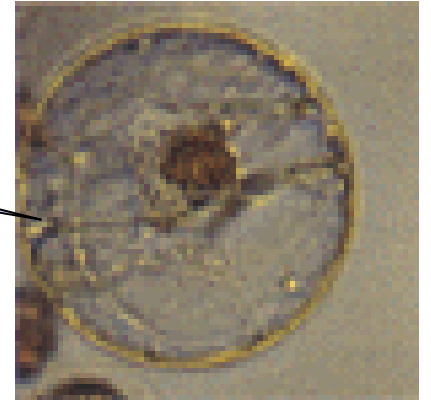
Amoeba
Proteus



Nerve
Cell



Bacteria



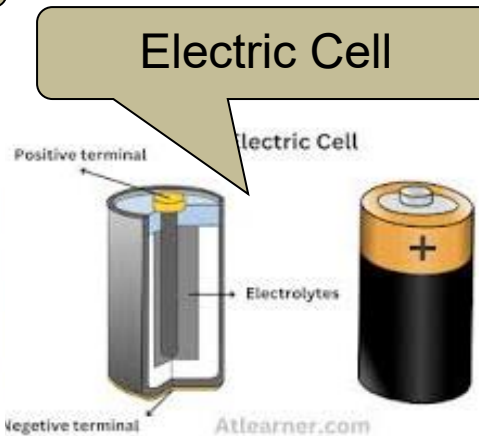
Plant Stem



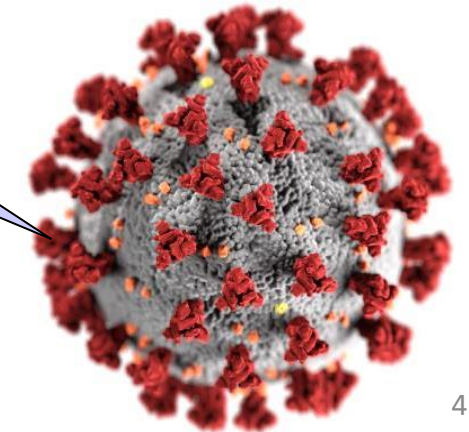
Red Blood
Cell



Battery Cell



Electric Cell



COVID-19
Virus

Cells are Two Types:

- Prokaryotic
- Eukaryotic

Life at cellular level

- **Robert Hooke (1665):** Hooke discovered a multitude of tiny pores that he named "cells".
- **Anton van Leeuwenhoek (1674):** He argued that motility was a sign of life, concluding that these were living organisms, and he went on to write numerous papers detailing various forms of microorganisms.
- **Schleiden (1838)** suggested that every structural part of a plant was made up of cells or the result of cells.
- **Theodor Schwann (1839)** states that along with plants, animals are composed of cells or the product of cells in their structure.
- After Schleiden and Schwann, **Rudolph Virchow (1855)** added the third tenets of cell theory.

Cell theory

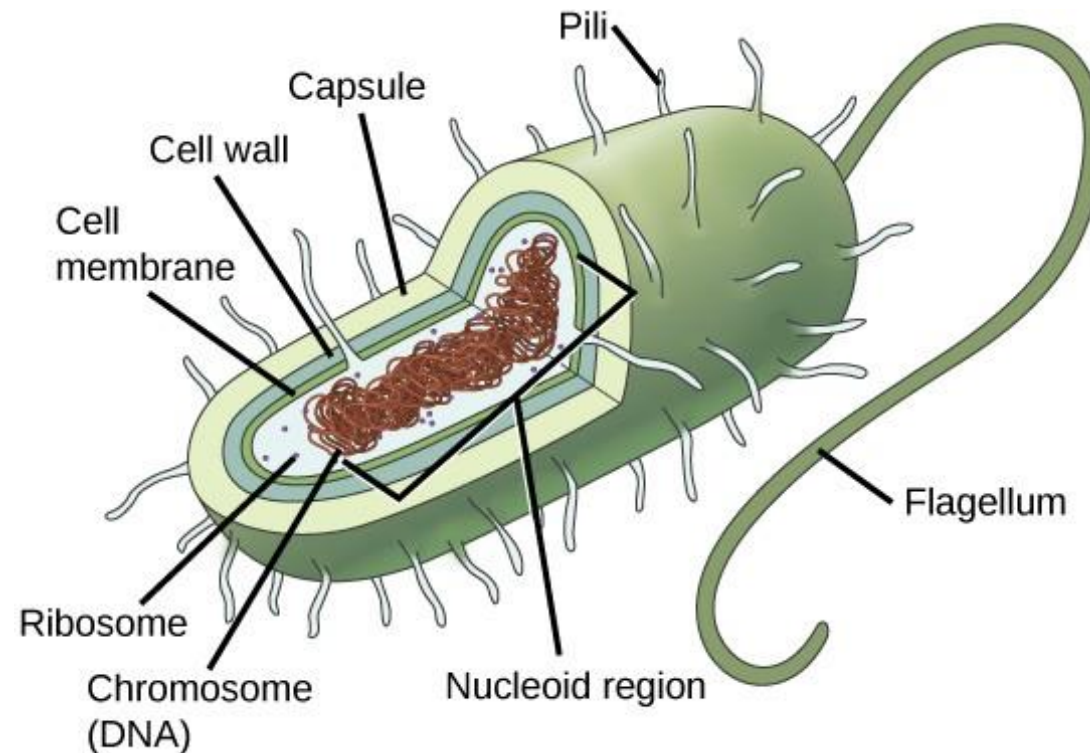
- Cells make up all living things.
- Cells are the basic building blocks for developing tissues, organs, and fully developed living creatures.
- Cells can only develop from other cells.

Modern Interpretation

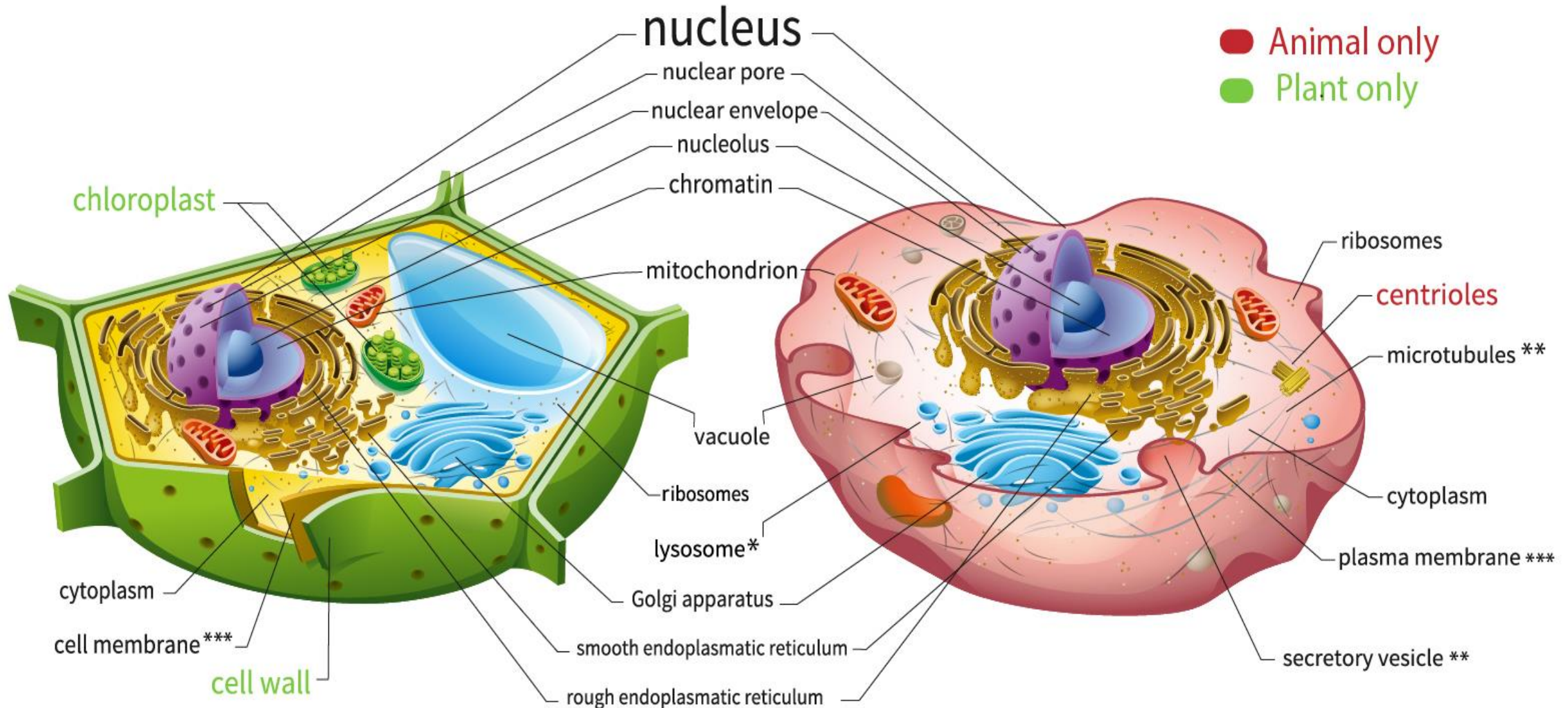
1. All known living things are made up of **one or more cells**
2. All living cells arise from pre-existing **cells by division**.
3. The cell is the fundamental **unit of structure and function** in all living organisms
4. **Energy flow** (metabolism) occurs within cells.
5. Cells **contain DNA** which is found in the chromosome and **RNA found in the cell nucleus** and cytoplasm.
6. All cells are basically the same in chemical composition in organisms of similar species.

Prokaryotic

- Do not have structures surrounded by membranes
- Few internal structures
- One-celled organisms, **Bacteria, virus, fungi, yeast, etc.**



Cell : Overview

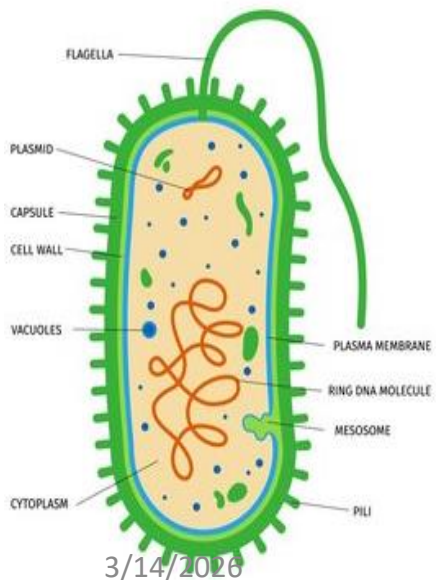


Cell : Overview

Cellular Structure

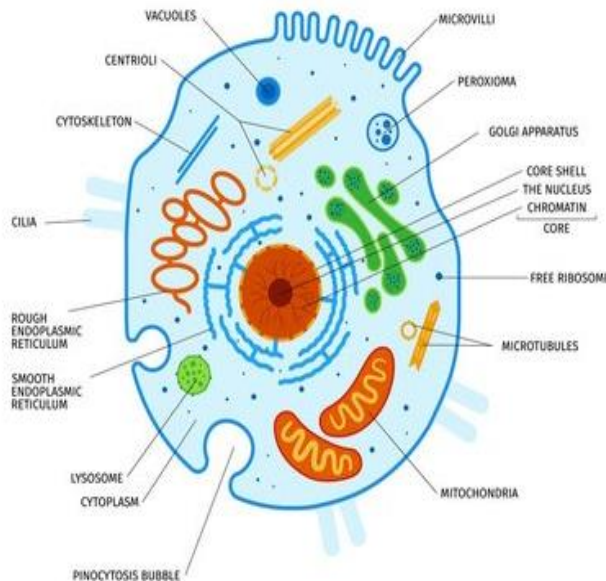
- Every living organism falls into one of two groups: **eukaryotes** or **prokaryotes**.
- Cellular structure determines which group an organism belongs to.

PROKARYOTIC CELL



3/14/2026

EUKARYOTIC CELL



Prokaryotic Cell

Unicellular

Lysosomes and
Peroxisomes absent

Microtubules absent

Endoplasmic reticulum
absent

Eukaryotic cell

Multicellular

Lysosomes and
Peroxisomes present

Microtubules present

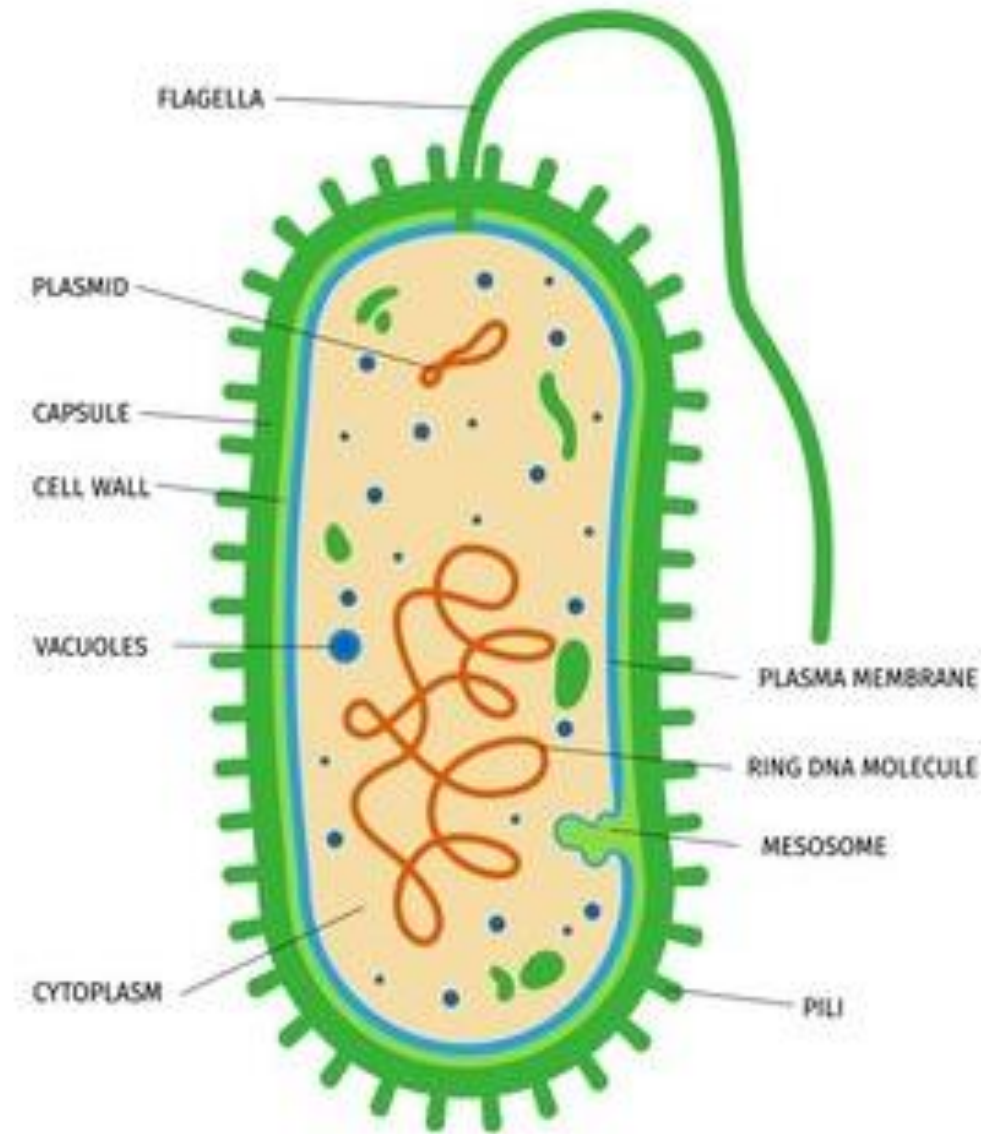
Endoplasmic reticulum
present

Prokaryotic Cells

- **Prokaryotic cells have no nucleus.** Instead, some prokaryotes such as bacteria have **nucleoid (nucleus like,** contains genetic materials).
- They all are **single-celled** microorganisms. Examples include archaea, bacteria etc.
- The cell size ranges from **0.1 to 0.5 μm .**
- The hereditary material can either be **DNA or RNA.**
- Prokaryotes generally reproduce by **binary fission,** a form of **asexual reproduction..**

Components of Prokaryotic Cells

- **Flagella:** Flagella are thin, tail-like structures that assist in movement
- **Plasmid:** A plasmid is a small circular DNA molecule found in bacteria and some other microscopic organisms.
- **Capsule:** Some bacteria have a layer of carbohydrates that surrounds.
- **Cell wall:** The cell wall provides structure and protection from the outside environment. Most bacteria have a rigid cell wall made from carbohydrates and proteins called peptidoglycans.

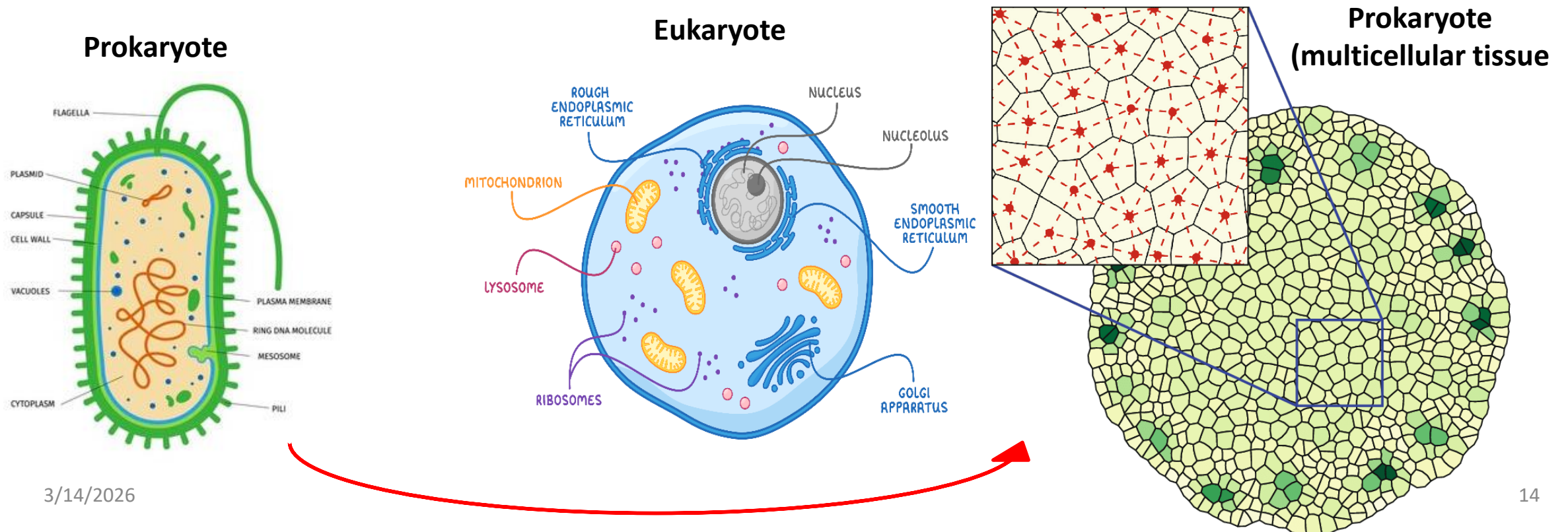


Components of Prokaryotic Cells

- **Vacuoles:** Vacuoles are membrane-bound organelles in prokaryotes that store nutrients, water, and waste products.
- **Cell membrane:** Every prokaryote has a cell membrane, also known as the plasma membrane, that separates the cell from the outside environment.
- **Nucleoid:** A central region of the cell that contains its DNA or RNA.
- **Mesosome:** help in the synthesis of the cell membrane, replication of DNA, and protein synthesis.
- **Pili:** Pili are rod-shaped structures involved in multiple roles, including attachment and DNA transfer.
- **Ribosome:** Ribosomes are responsible for protein synthesis.
- **Capsule:** Some bacteria have a layer of carbohydrates that surrounds the cell wall called the capsule. The capsule helps the bacterium attach to surfaces.
- **Fimbriae:** Fimbriae are thin, hair-like structures that help with cellular attachment

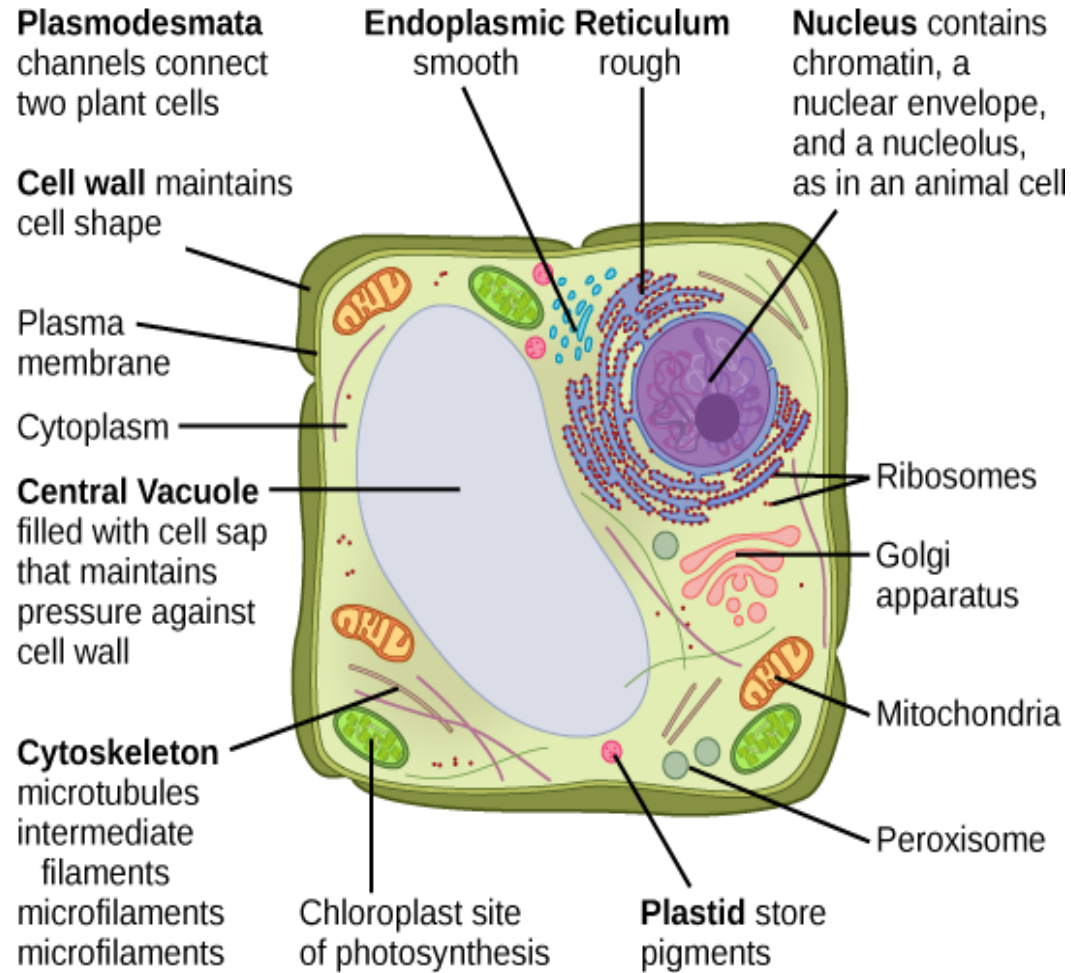
Eukaryotic Cells

- Contain organelles surrounded by membranes
- Most multicellular organisms/plants/animals

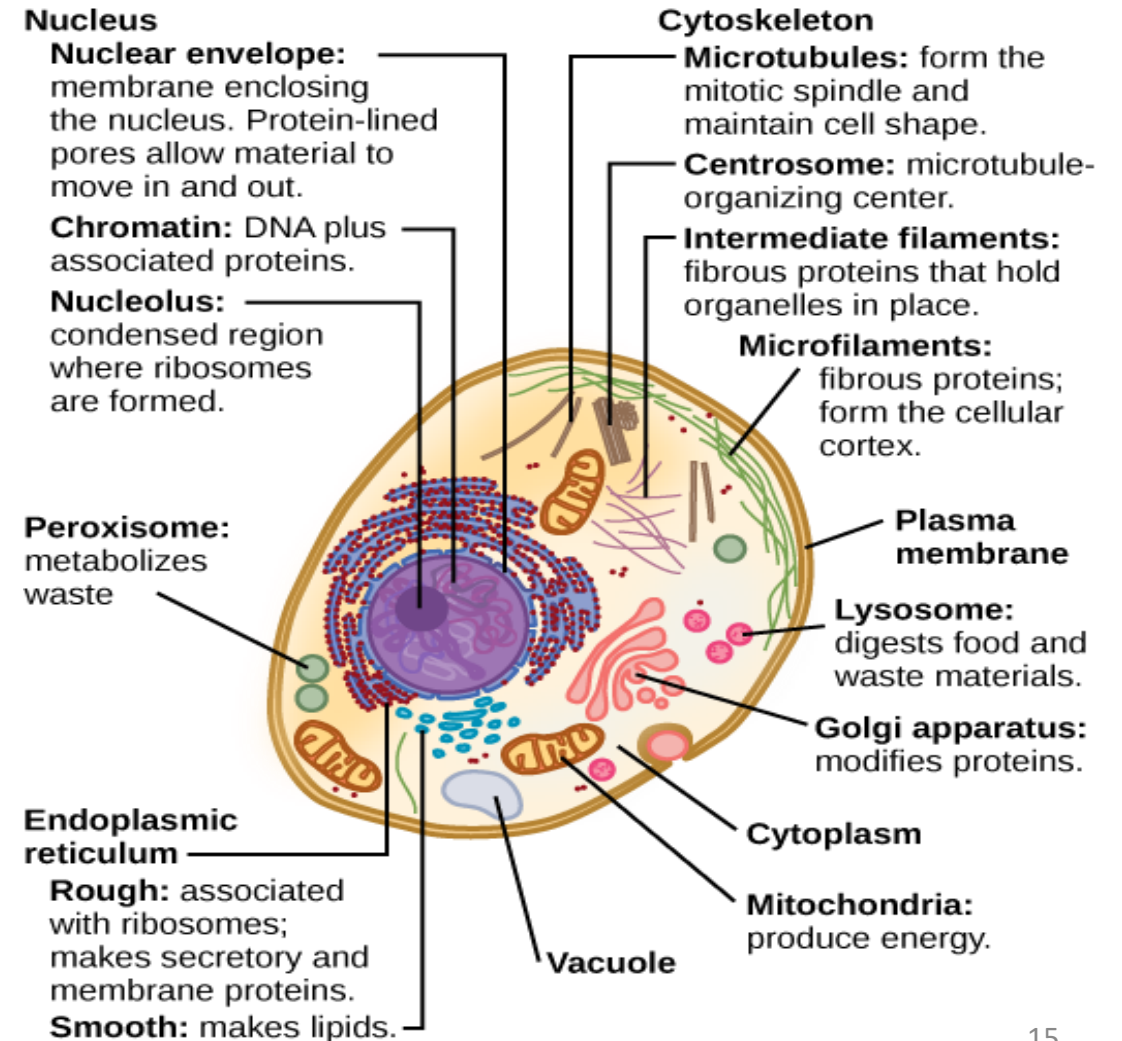


Eukaryotic Cells

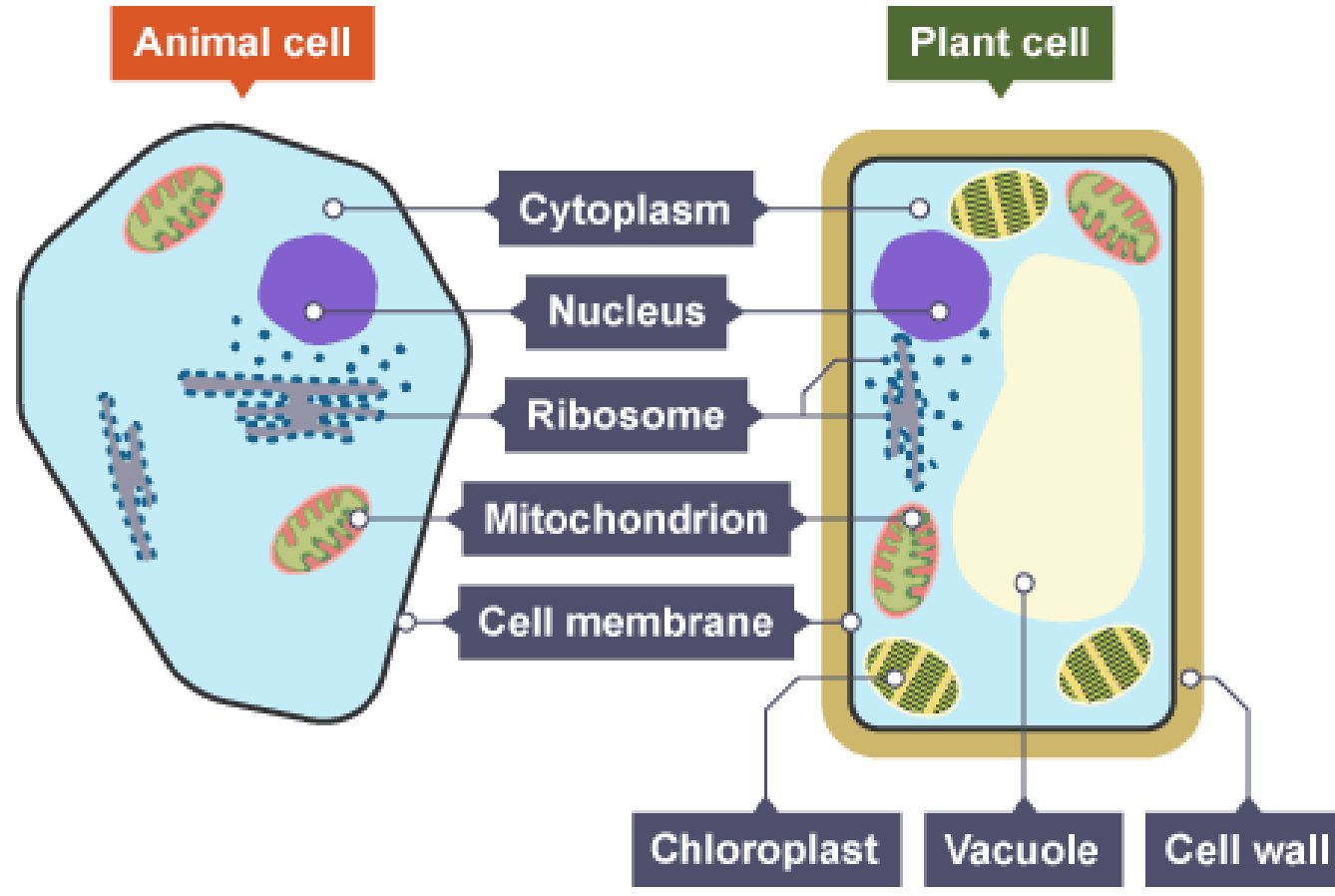
“Typical” Plant Cell



“Typical” Animal Cell



Key difference in Plant and Animal Cells



There are some contrasting features between plant and animal cells. For e.g., the plant cell contains **chloroplast**, **central vacuoles**, and other **plastids**, whereas the **animal cells do not**.

Eukaryotic Cells

- Eukaryotic cells are characterized by a **true nucleus**.
- The size of the cells ranges between **10–100 μm** in diameter.
- This broad category involves **plants, fungi, protozoans, and animals**.
- The plasma membrane is responsible for the **transport of nutrients** and electrolytes **in and out** of the cells. It is also responsible for **cell-to-cell communication**.
- They reproduce **sexually as well as asexually**.

Cell wall

Cell wall: It is an outer non living, rigid layer of cell. It is present in bacterial cells, fungal cells and plant cells. It is a permeable membrane chiefly composed of cellulose. It gives rigidity, mechanical support and protection to the cell.

Question: Do all cells have a cell wall?

Answer: Not all cells have cell walls.

Question: Which cells, therefore, have a cell wall?

Answer: Prokaryotes -except mycoplasma and L-form bacteria- and some eukaryotes have a cell wall.

Question: Do animal cells have a cell wall?

Answer: Cell walls are not present in animal cells. That's why animal cells are not as rigid as other cells surrounded by cell walls. Animal cells are, therefore, have more flexibility than plant cells.

Cell membrane

Cell membrane: The cell membrane, also known as the plasma membrane, is a semipermeable lipid bilayer that separates the cell's interior from its exterior environment. It's a dynamic, and fluidic in structure.

- **Functions:** It allows the outward and inward movement of molecules across it. The movement of molecules across the plasma membrane takes place by diffusion, osmosis, active transport, phagocytosis (cell eating) and pinocytosis (cell drinking).

Cell membrane

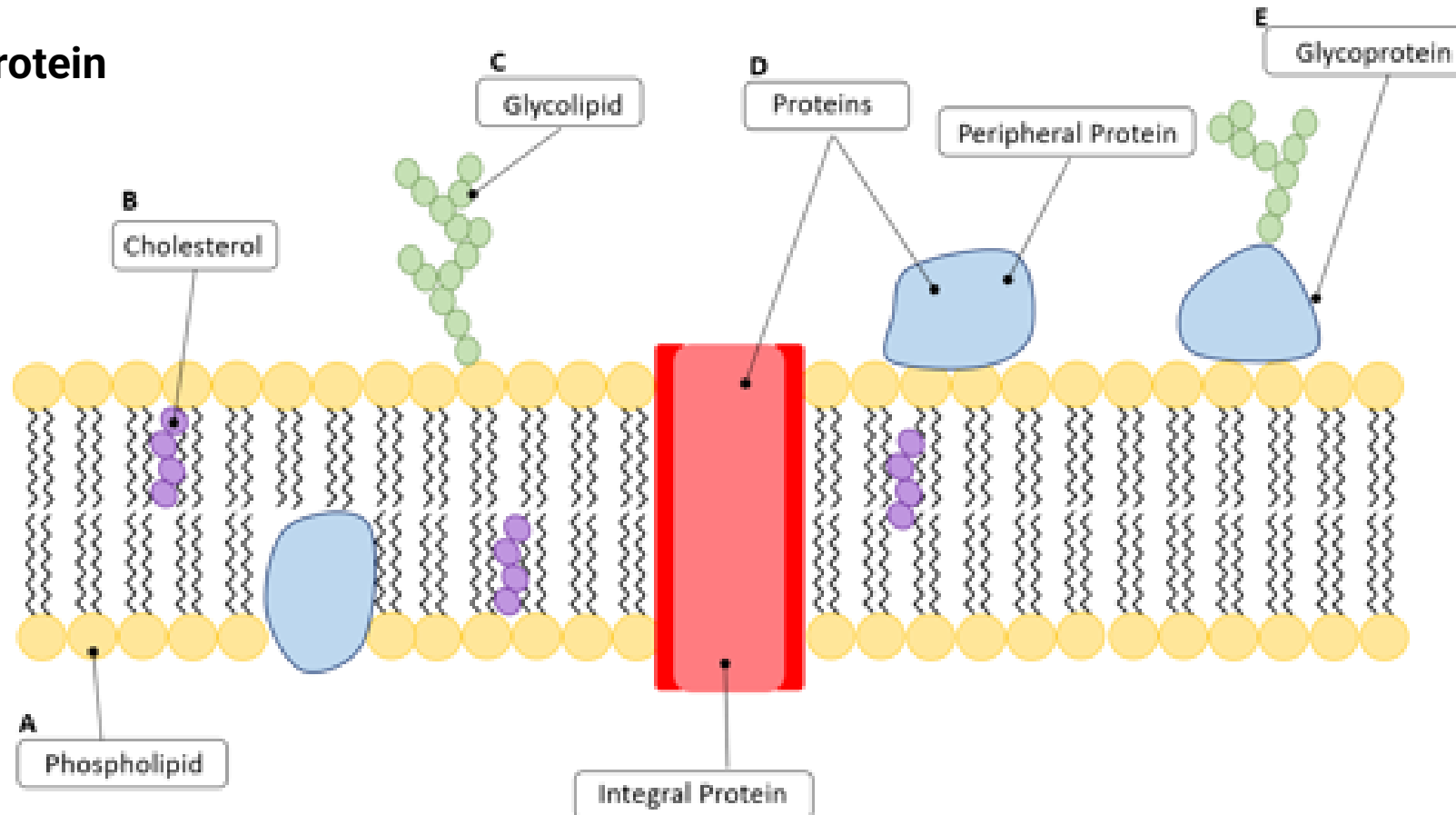
- Lipid bilayer is made of the following:

- 3 types of proteins:

- Integral proteins
- Peripheral proteins
- Glycoprotein

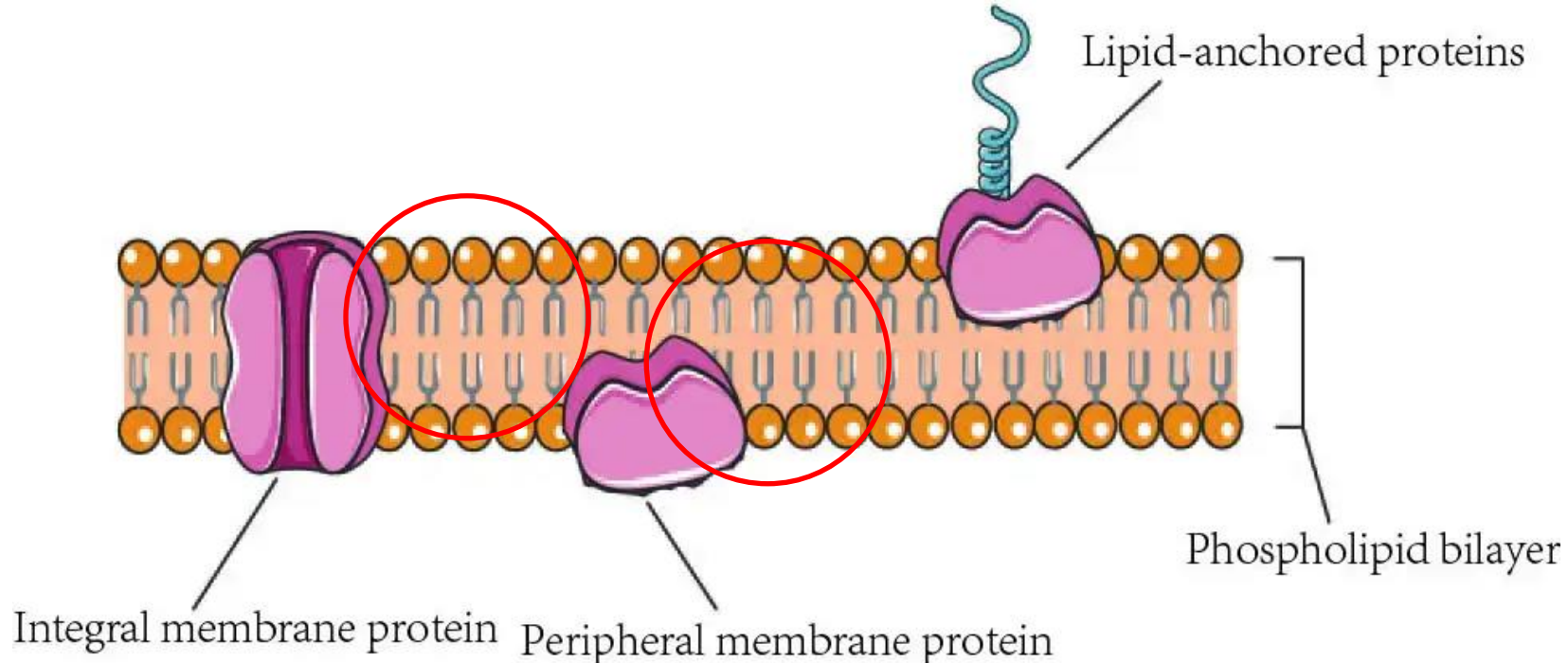
3 types of lipids:

- Membrane Phospholipids
- Membrane glycolipids
- Cholesterol



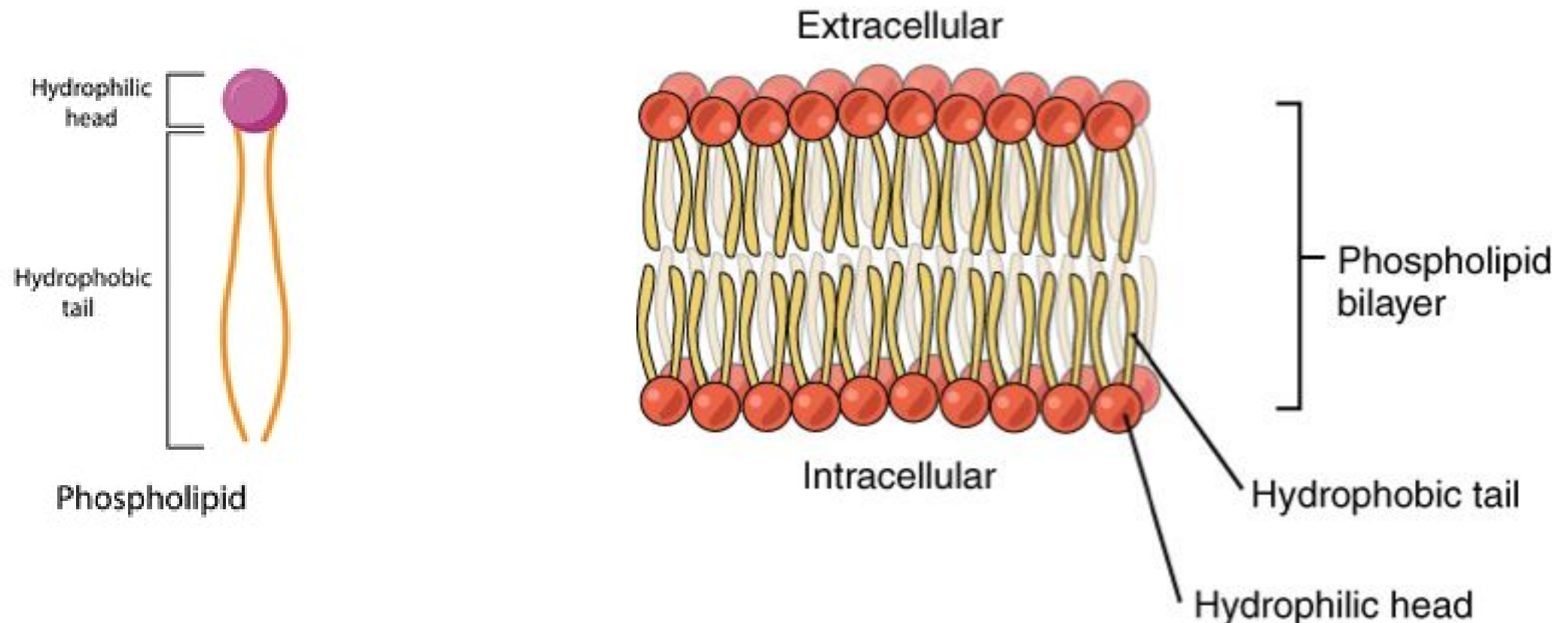
Membrane proteins

- **Integral proteins** (Transmembrane proteins)- spanned throughout the bilayer. Integral proteins are amphipathic, have both hydrophobic and hydrophilic regions.
- **Peripheral proteins**- linked at the cytoplasmic surface by attachment to a fatty acid chain and linked at the external cell surface attached by an oligosaccharide.
- **Lipid anchored proteins** and **glycoproteins** are also present in membrane.



Membrane Lipids : Phospholipid

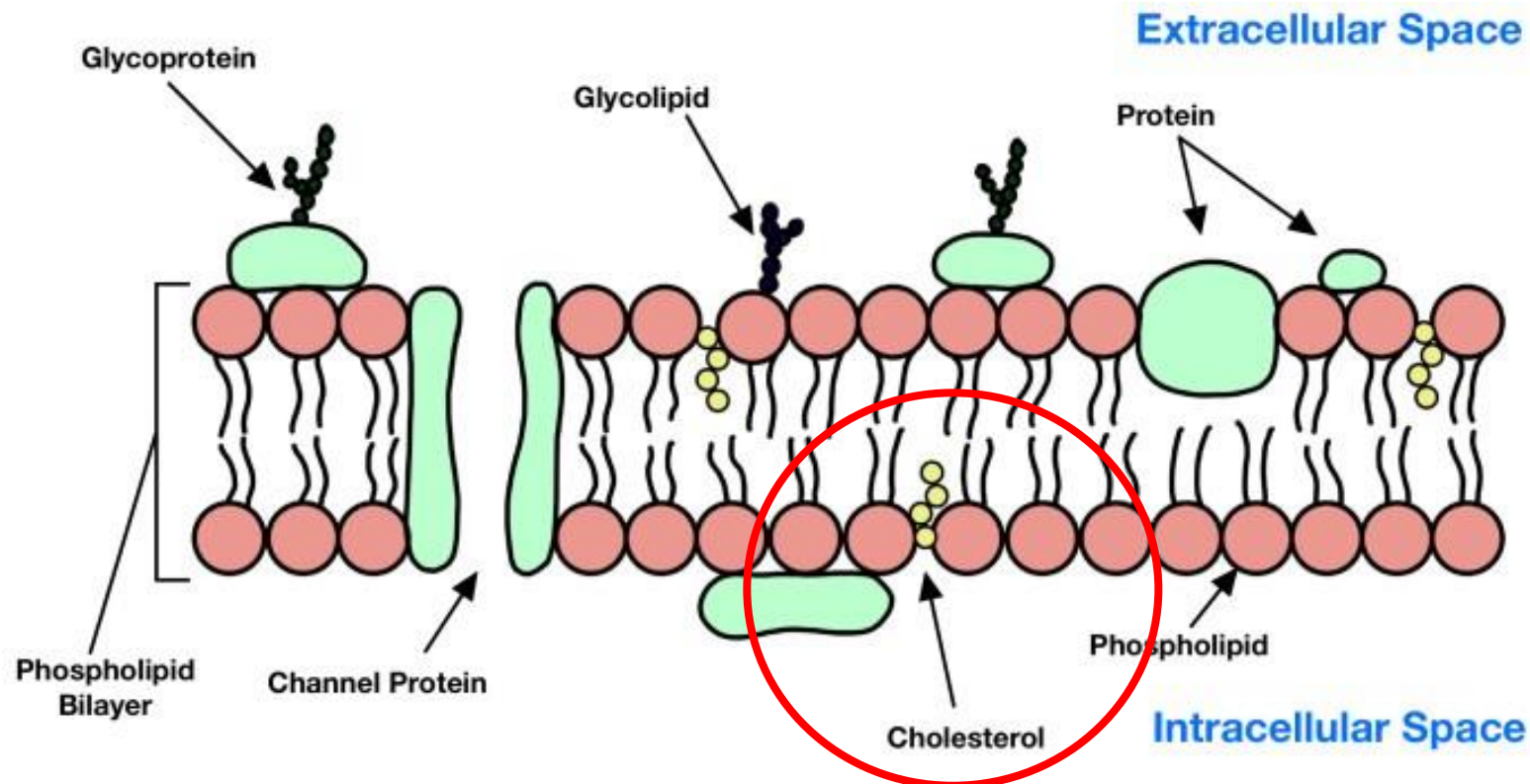
- Phospholipids are amphipathic molecules that form a bilayer in cell membranes.
- The fatty acids have hydrophobic hydrocarbon chains that face the interior of the membrane, while the polar head groups face the outside



Membrane Lipids : Cholesterol

- The amount of cholesterol may vary with the type of membrane.
- Plasma membranes have nearly one cholesterol per phospholipid molecule.

Function: This makes the lipid bilayer less deformable. Without cholesterol (such as in a bacterium) a cell would need a cell wall. Also keeps the cell membrane from becoming too stiff.



Cell Structure and Function

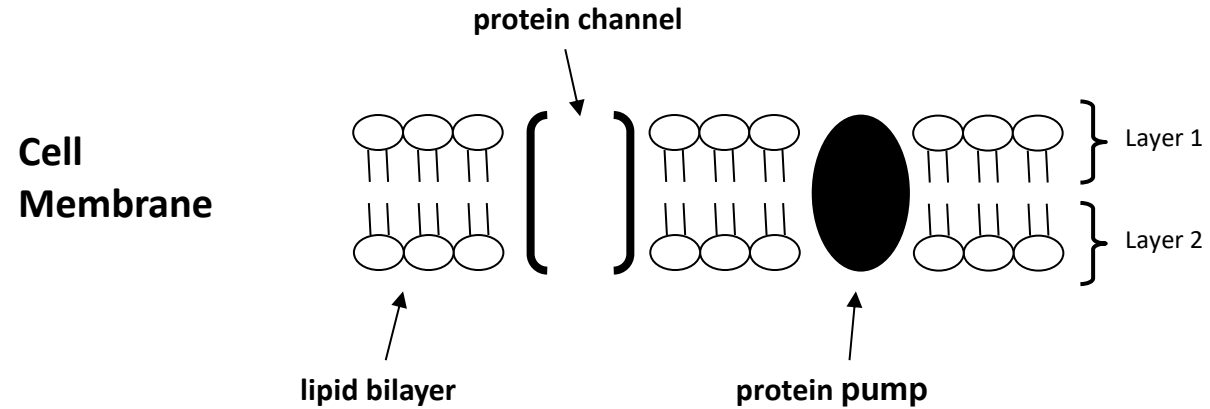
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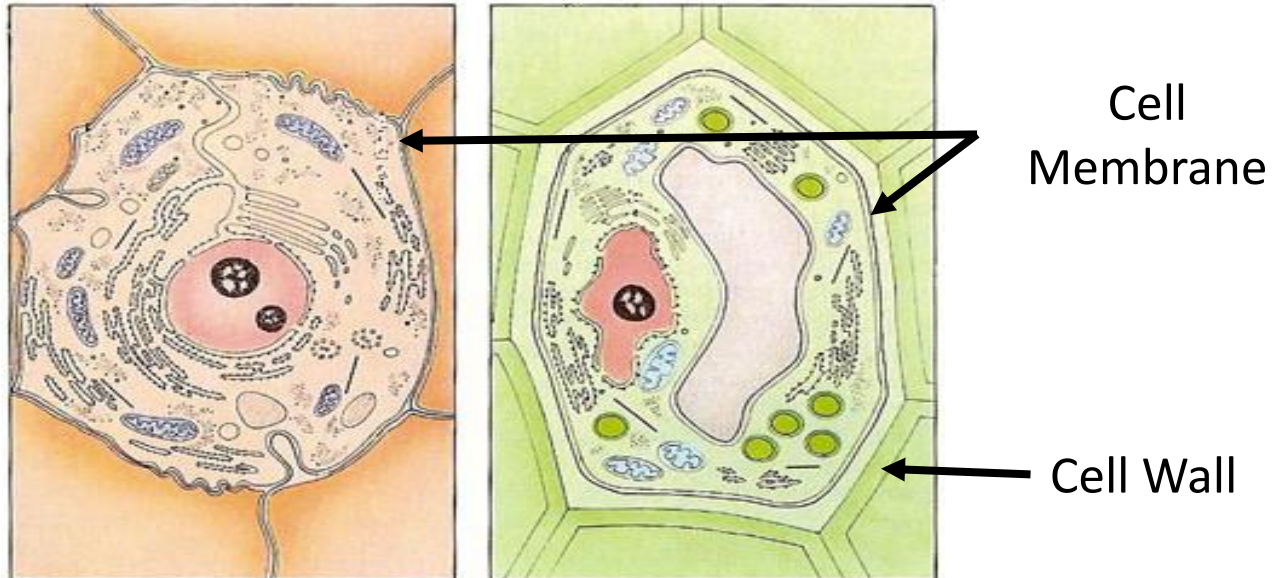
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Cell Membrane and Cell Wall:

ALL cells have a cell membrane made of proteins and lipids



SOME cells have cell membranes **and** **cell walls** – ex: plants, fungi and bacteria



Cell Wall vs Cell Membrane

Cell wall	Cell membrane
The outermost layer, next to the cell membrane	A bilipid layer surrounding the cell contents, such as cytosol and organelles
Typically, 0.1 μm to several μm in thickness	Typically, 7.5–10 nm in thickness
Components vary depending on the species	Made up of lipid bilayer, with carbohydrates and lipoproteins
Confers rigidity, giving the cell a more definite shape	With only a cell membrane, the cell is more flexible and can change shape as needed
Confers protection, e.g. against the impact of osmotic pressure; more permeable to small molecules	Confers protection, e.g. by being selective, regulating the passage of substances, and so, not all can readily enter the cell.
Lacks cell surface receptors	Have cell surface receptors
Present in plants, fungi, protists (e.g. algae and molds), and bacteria	Present in all cells

Movement Through the Membrane

- Materials can move through the membrane by:
 - **Passive Transport/ Diffusion** (No energy required)

[high] → [low]

- Osmosis
- Facilitated Diffusion

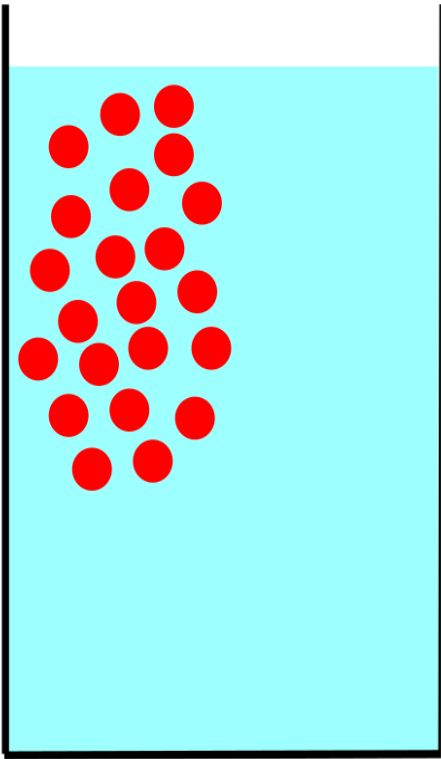
- **Active Transport** (Energy required)

[low] → [high]

- Protein Pumps
- Endocytosis
- Exocytosis

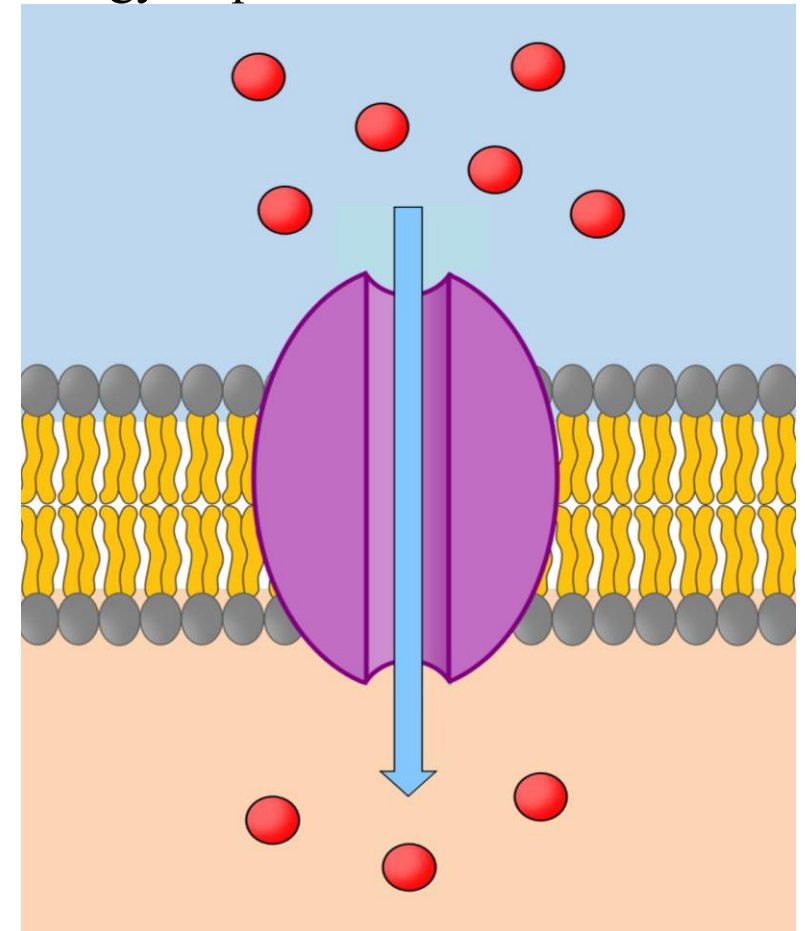
Diffusion

- Requires **no energy (ATP)**
- Moves from an area of **High concentration → low** concentration until dynamic equilibrium is reached.



Facilitated Diffusion

- Diffusion **with the help of transport** proteins
- No energy required



Osmosis

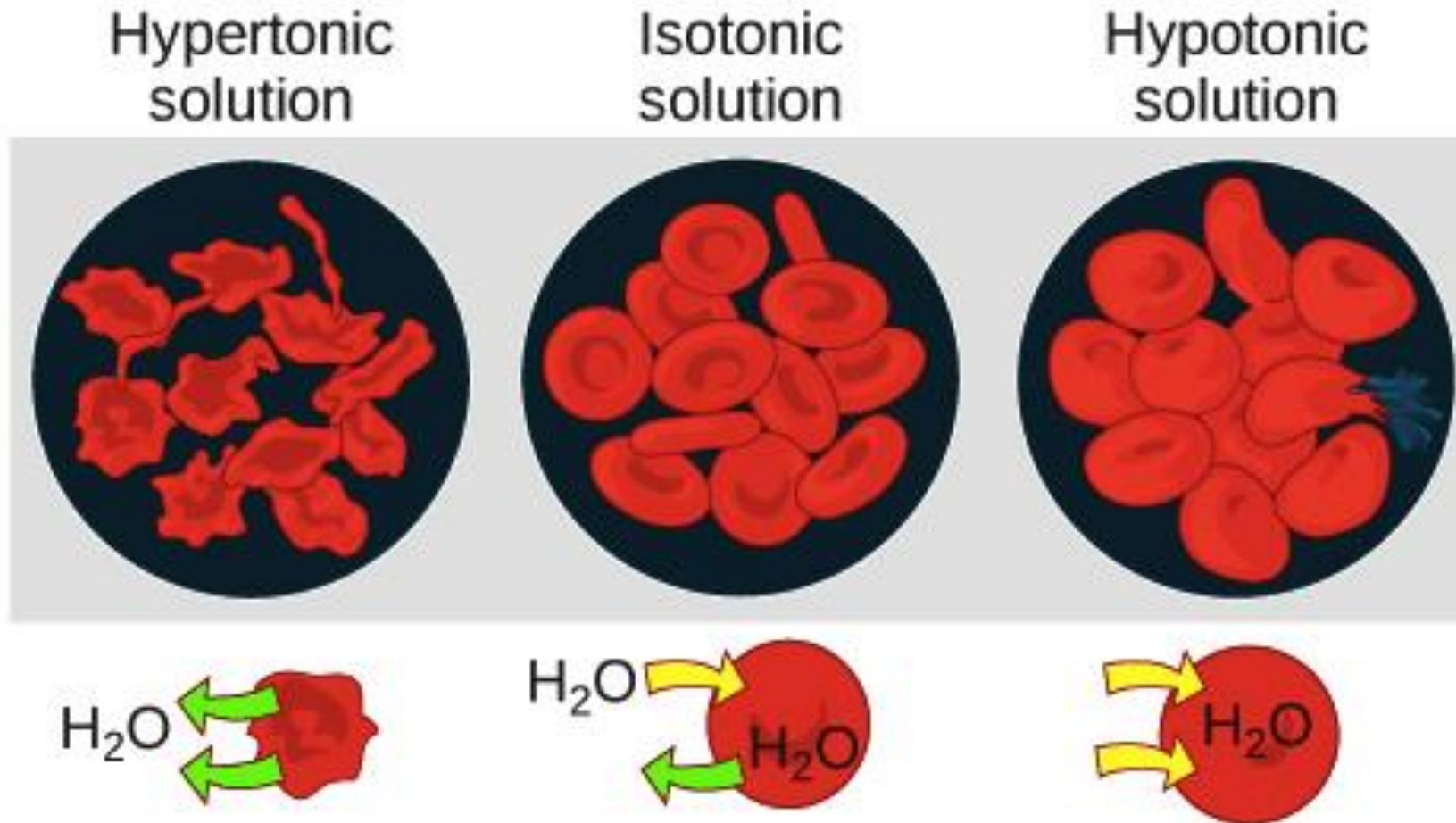
- A type of diffusion (no energy needed)
- Allows water molecules to pass easily through the selectively permeable membrane.
- Solution = solute + solvent
 - Solute = sugar (or another dissolved substance)...CANNOT go through the membrane
 - Solvent = water...CAN go through the membrane

ONLY water moves

- The solute stays put on one side or the other
- Water moves back and forth according to the concentration of water on each side of the membrane.

Osmotic Pressure

- **Hypertonic solutions:** One solution has more solute and less water compared to the other solution.
- **Isotonic solutions:** The 2 solutions have equal concentrations of solute and solvent.
- **Hypotonic solutions:** One solution has less solute and more water compared to the other solution.



Transport through cell membrane

Transport Concepts



Passive

No energy needed

Passive
vs.
Active



Active

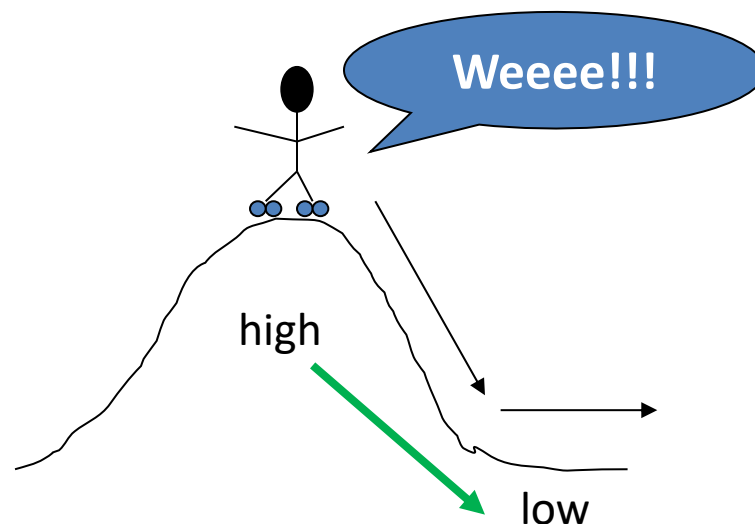
Energy needed

Types of Cellular Transport

Passive Transport

cell doesn't use energy

1. Diffusion
2. Facilitated Diffusion
3. Osmosis

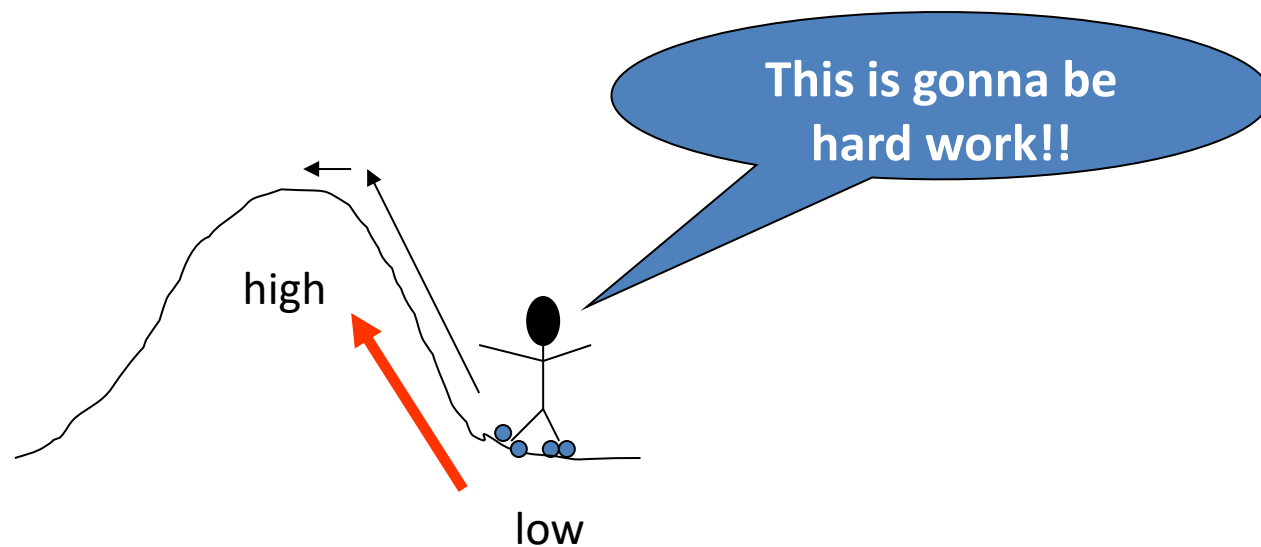


• Animations of Active Transport & Passive Transport

Active Transport

cell does use energy

1. Protein Pumps
2. Endocytosis
3. Exocytosis.



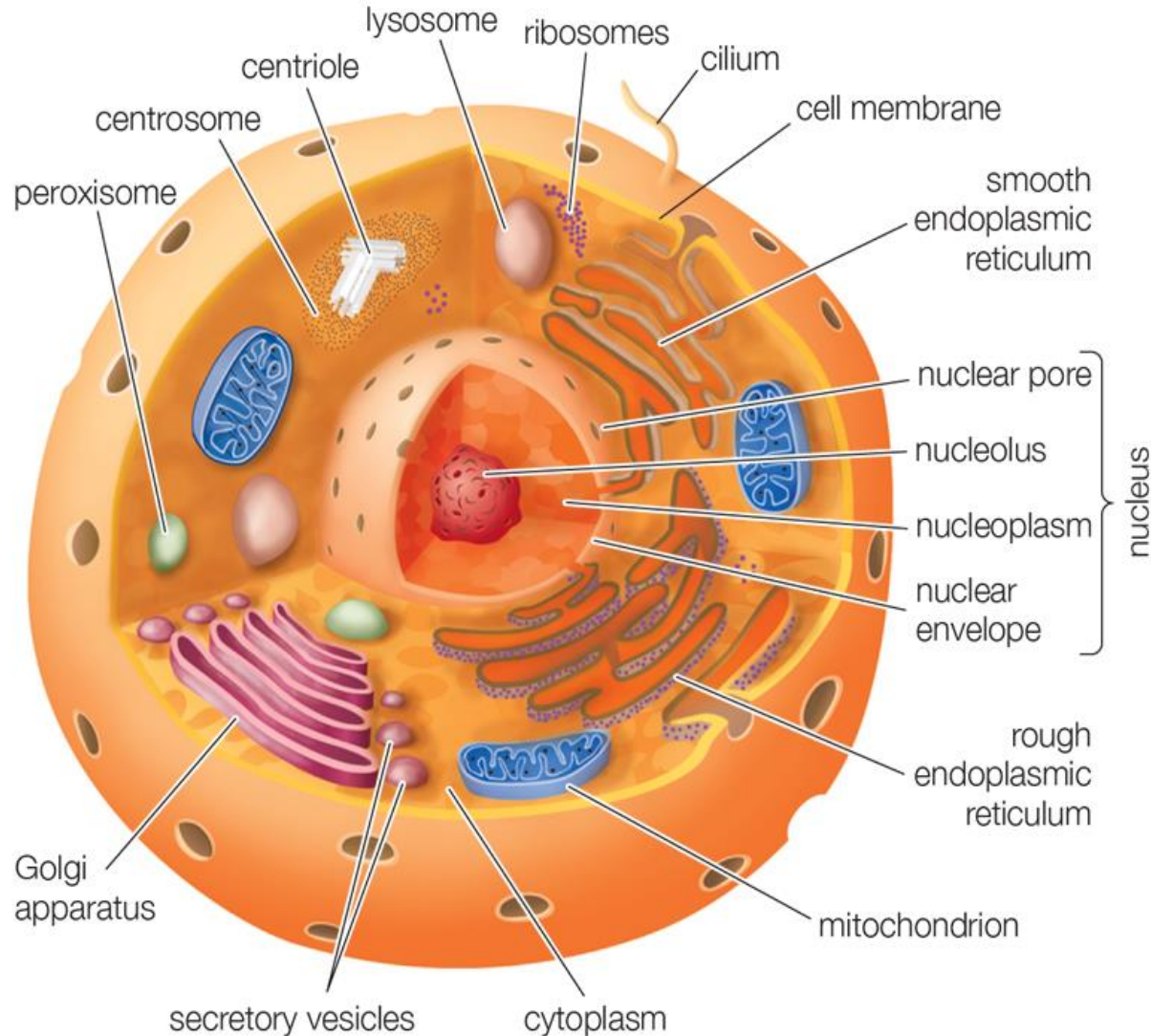
Cell components: **Protoplasm**

- **Protoplasm:** It is a living substance of the cell that possesses all vital products made up of inorganic and organic molecules. It includes **cytoplasm and nucleus**. **Purkinje (1837)** coined the term protoplasm. **Huxley** called protoplasm as “physical basis of life”.
- **Cytoplasm:** It is the **jellylike, semi fluid matrix present between the cell membrane and nuclear membrane**. It has various living cell inclusions called cell organelles and non living cell inclusions called ergastic substances and cytoskeletal elements. The cytoplasm without cell organelles is called cytosol.

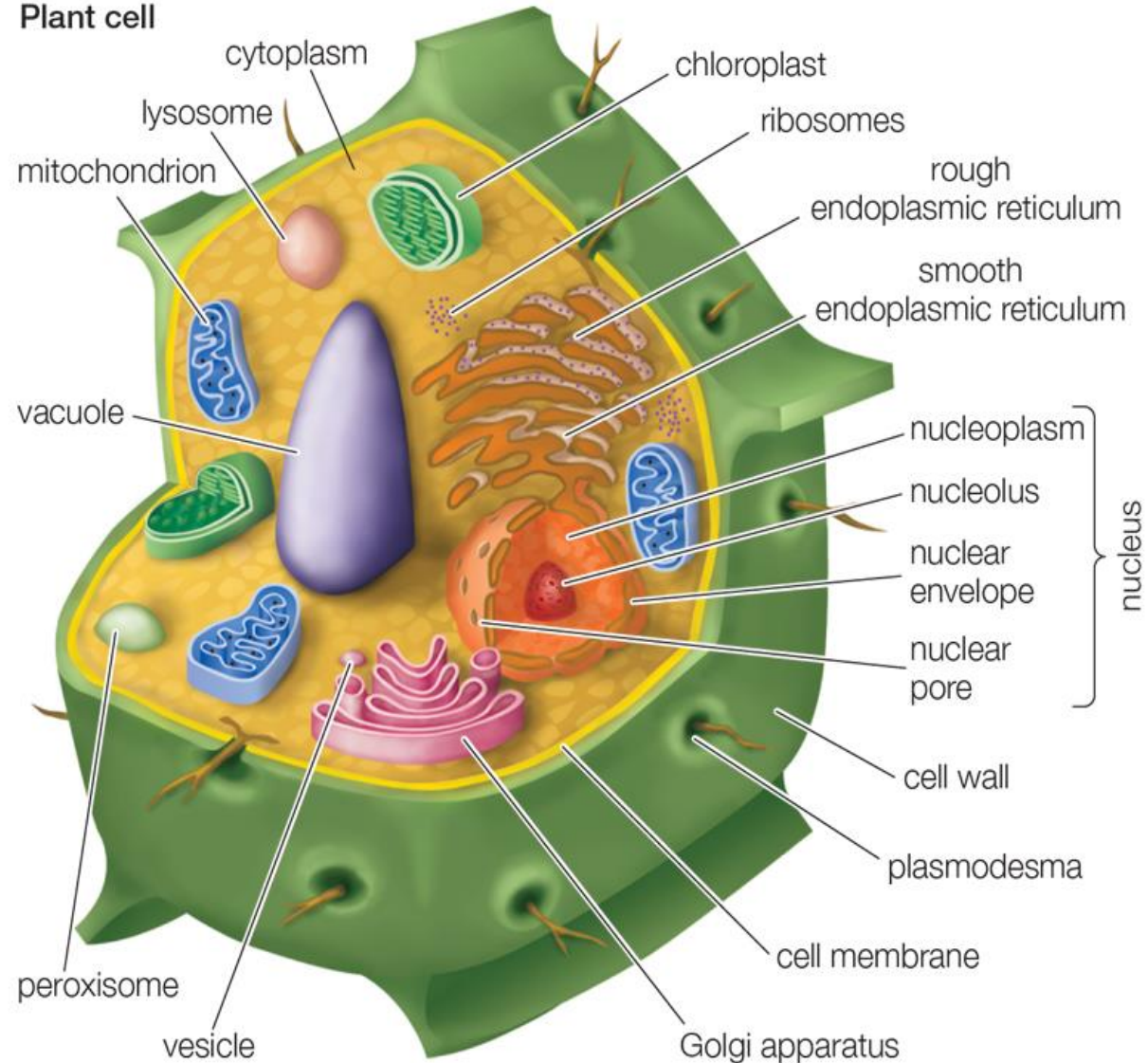
Cell components: Protoplasm

Typical animal cell and plant cell

Animal cell



Plant cell



Cell components: **Cytoplasm**

Cytoplasm is a **gel-like fluid that fills the inside of a cell** and performs many functions, including:

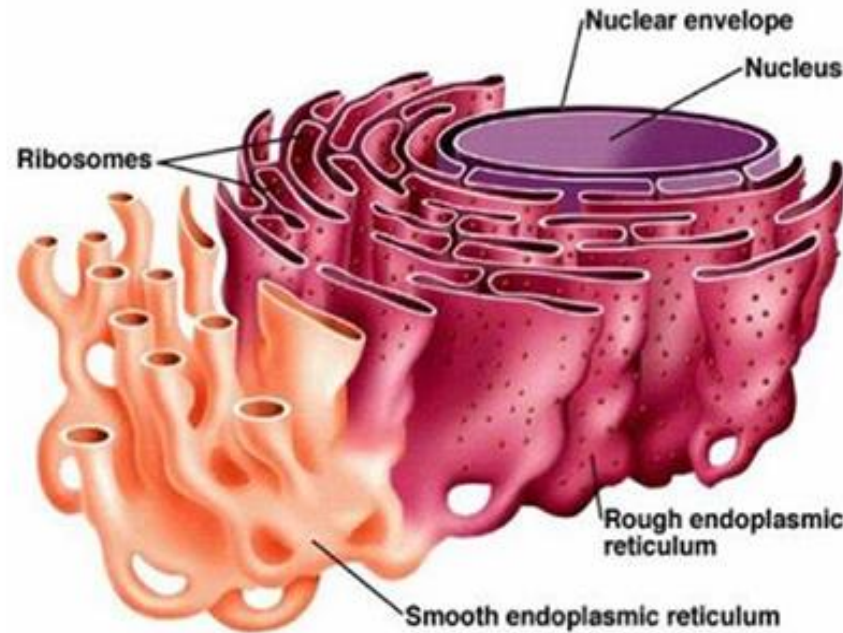
- Provides a **structure for the cell parts** to move around within the cell membrane.
- Allows **materials to move between organelles through diffusion**.
- **Stores the cell's genetic material** and products of cellular respiration.
- **Protects the cell.**
- **Hosts metabolic processes and contains enzymes** that break down waste and aid in metabolic activity.
- **Maintains the cell's shape.**
- **Contains ions that help maintain osmotic balance** inside the cell.

Cell components: Endoplasmic reticulum

Membrane bound cell organelles present in cytoplasm.

Structure

1. It is an irregular network of double membranes tube-like structure distributed over the entire cytoplasm in a cell.
2. It is continuous with the cell or plasma membrane on the outside and nuclear membrane of the inside.
3. At its outer end, endoplasmic reticulum (ER) connects itself with the cell membrane.
4. At the inner end, it is connected with the nuclear membrane.
5. It is of two types – rough and smooth endoplasmic reticulum. It appears rough when the particles-like ribosomes are attached to it and appear smooth without them.

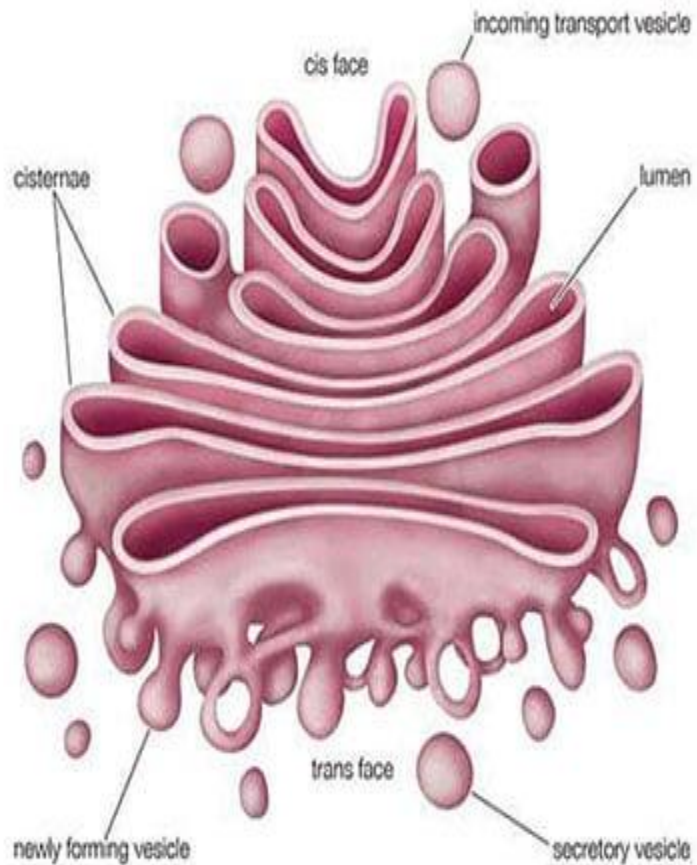


Function

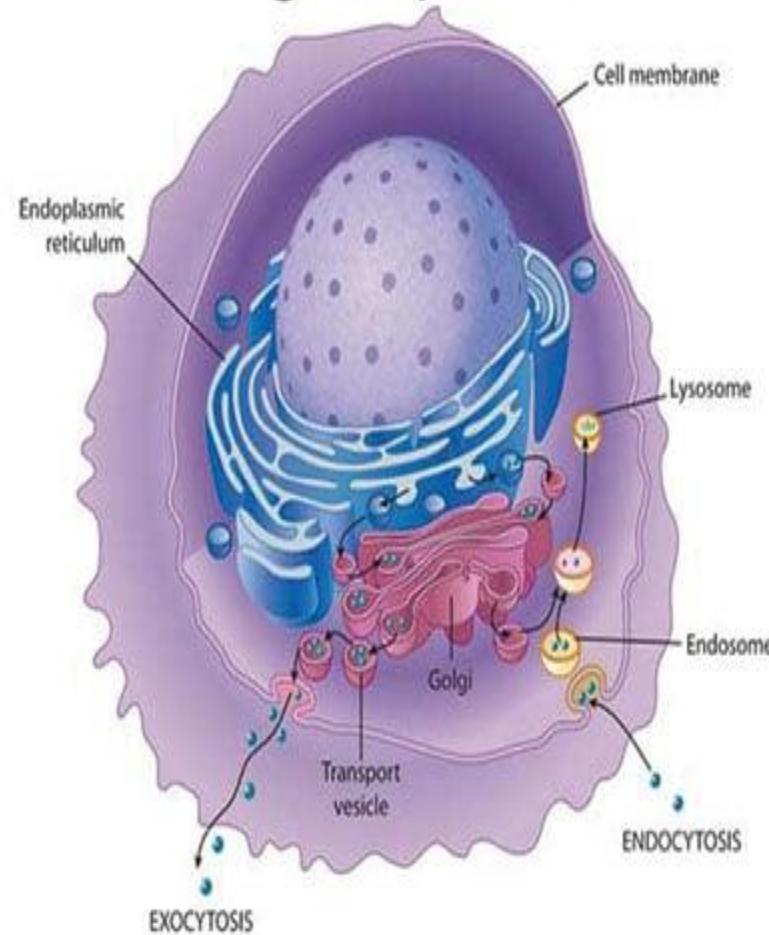
1. Its main function is to form the supporting framework of the cell.
2. It also serves as a pathway for the distribution of the material (protein and fats) from one part of the cell to the other.
3. Due to the presence of ribosome - proteins and lipids are synthesized on the rough endoplasmic reticulum, which helps in regenerating cell membranes. This process is known as membrane biogenesis. ('Biogenesis' mean 'generation of a substance by living matter').

Cell components: Golgi Bodies

Golgi Apparatus



Golgi Body in a Cell



Golgi complex has a group of curved, flattened plate like compartments called cisternae. They stacked one above the other like pancakes. The cisternae produce a network of tubules from the periphery. These tubules end in spherical enzyme filled vesicles.

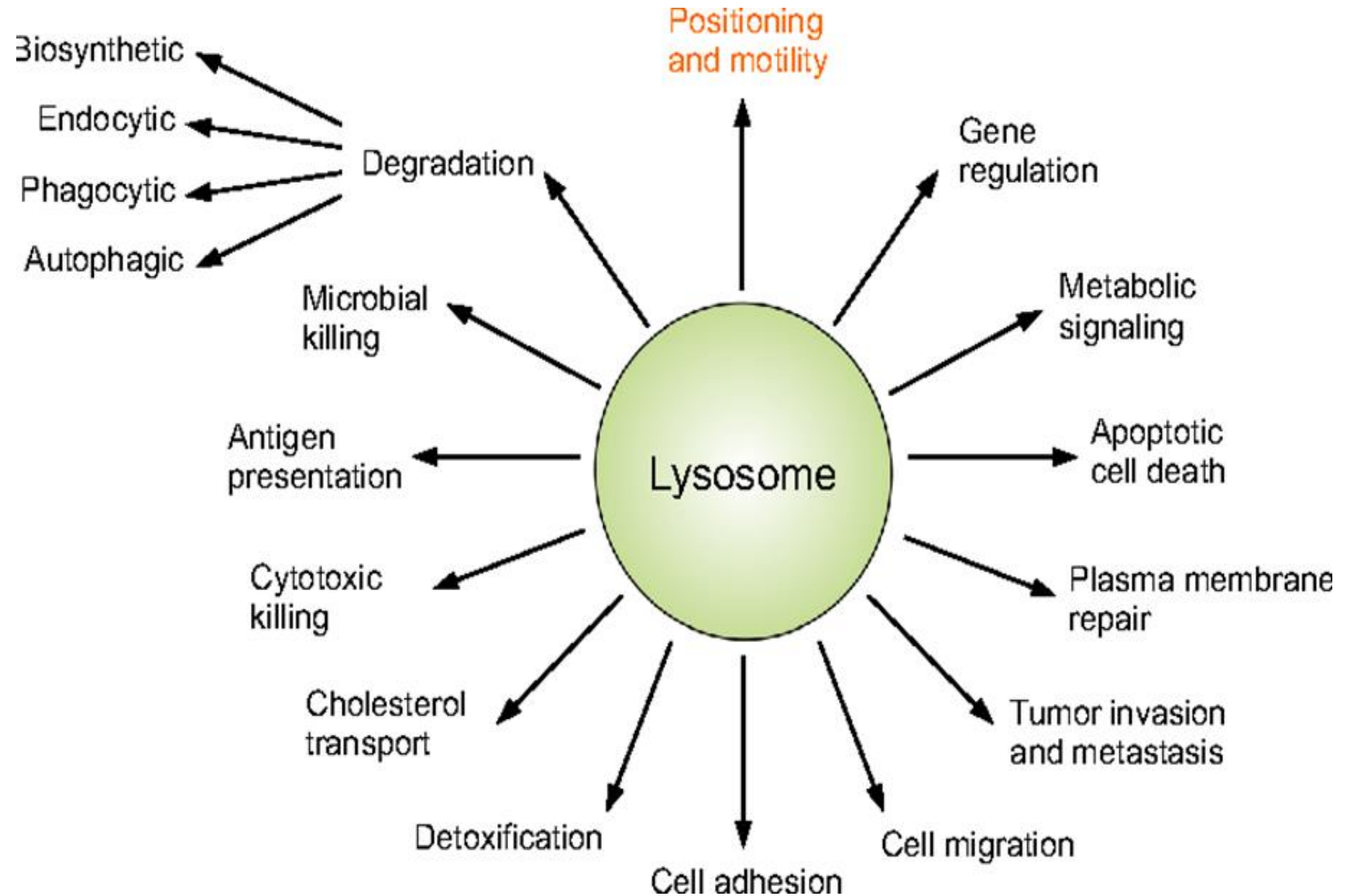
Common name: "**Packaging centers**" of the cell

Functions:

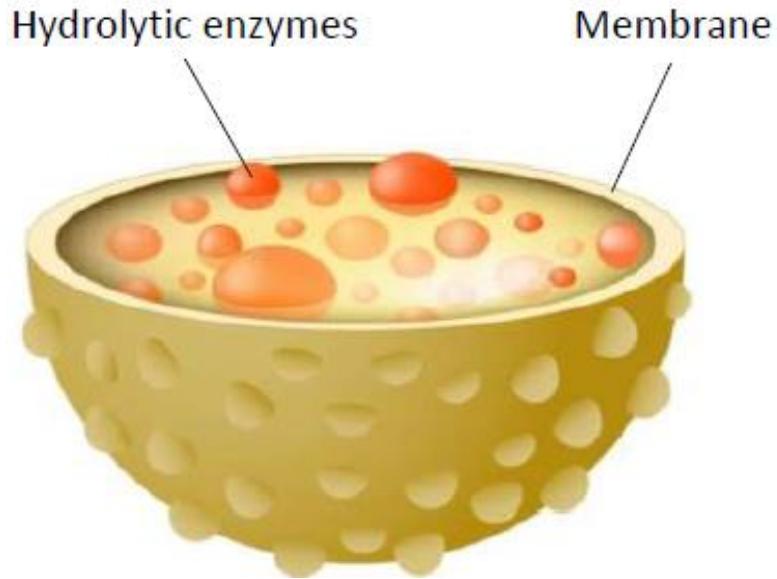
- **They pack enzymes, proteins, carbohydrates etc. in their vesicles, hence called packaging centers.**
- **They produce Lysosomes.**
- **They secrete various enzymes, hormones and cell wall material**

Cell components: Lysosomes

Autolysosomes:
Suicidal bags of cell /
Time bombs of the cell /
Recycling centers.



Cell: Lysosome



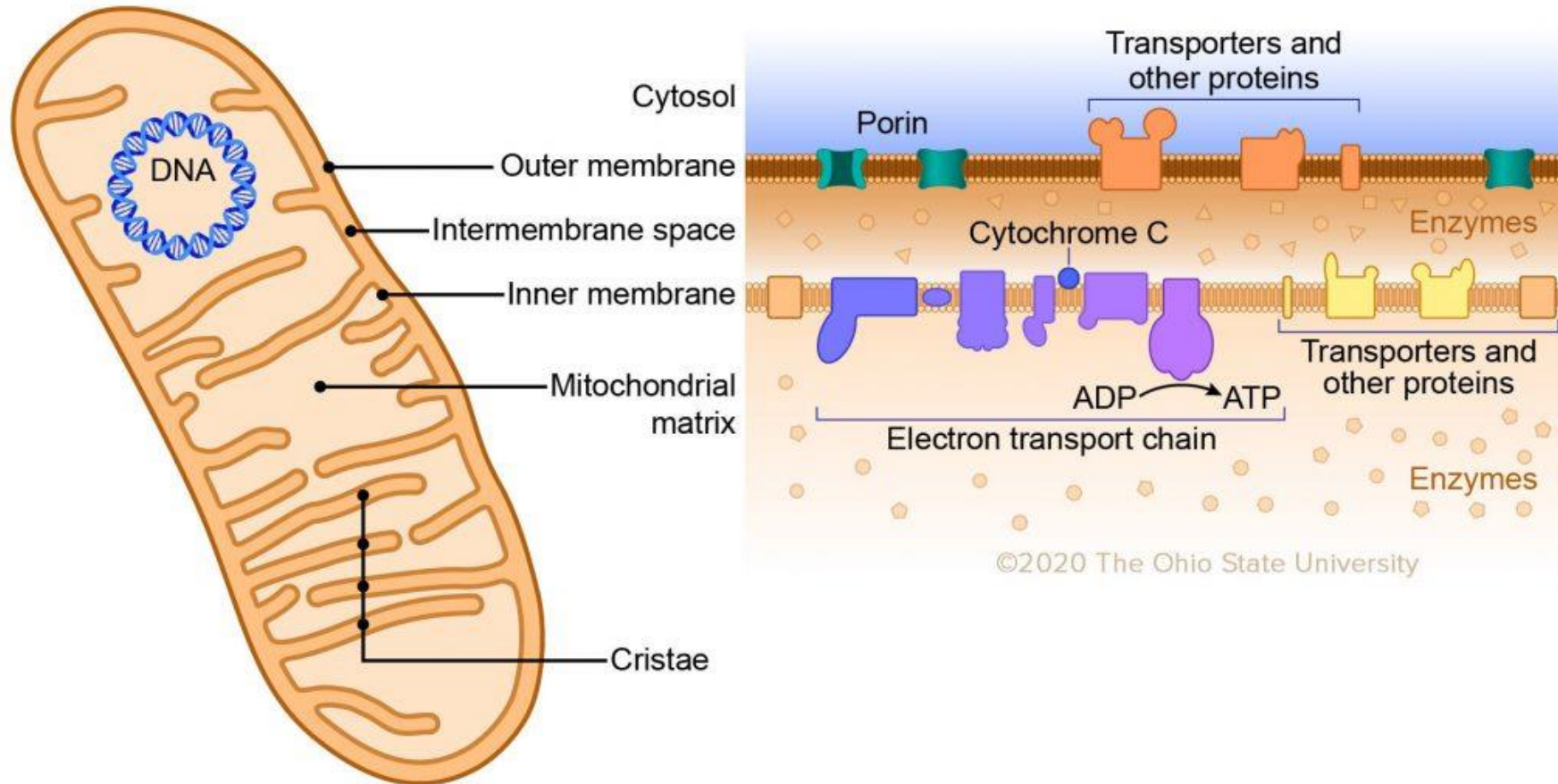
- Small, spherical, single membrane sac
- Found throughout the cytoplasm
- Filled with hydrolytic enzymes
- Occur in most animal cells and in few type of plant cells

Functions:

- Help in digesting of large molecules
- Protect cell by destroying foreign invaders like bacteria and viruses and other pathogens
- Degradation of worn-out organelles
- In dead cells perform autolysis

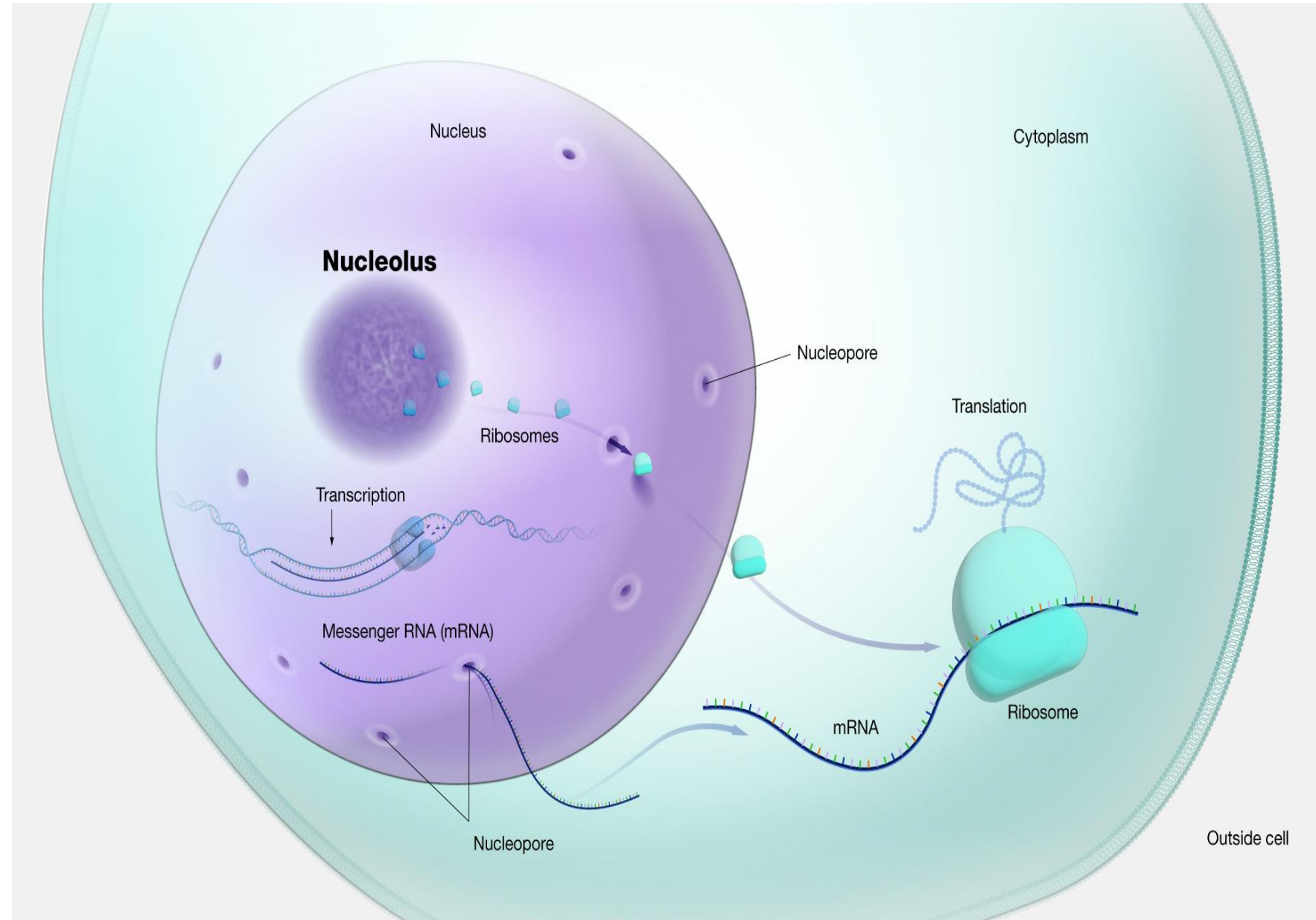
Cell components: Mitochondria

- **Kolliker** (1880)- discovered in the muscle cells of insects, **Altman** called them as **Bioplasts**, **Benda** (1897) coined the term **Mitochondria**
- **Common name: Power houses of the cell / Storage batteries of the cell**
- **Functions:** Mitochondria **synthesise and store the energy rich molecules ATP** (Adenosine triphosphate) during aerobic respiration. So, they are called “**Power houses of the cell**”.

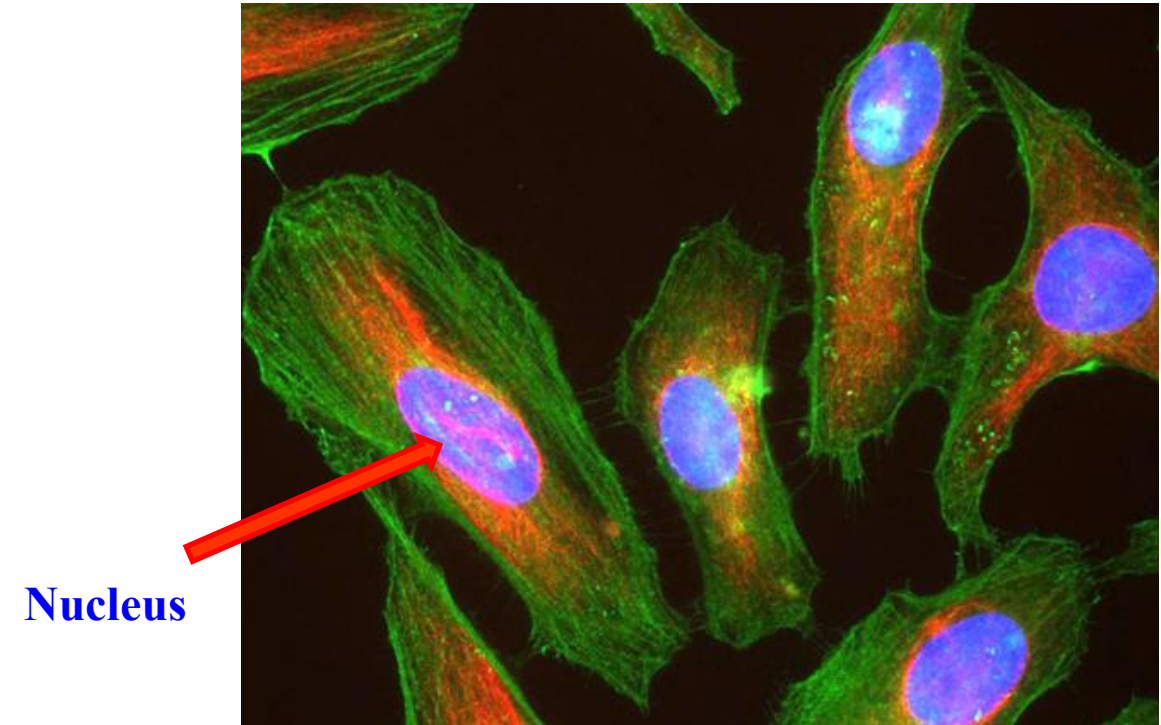
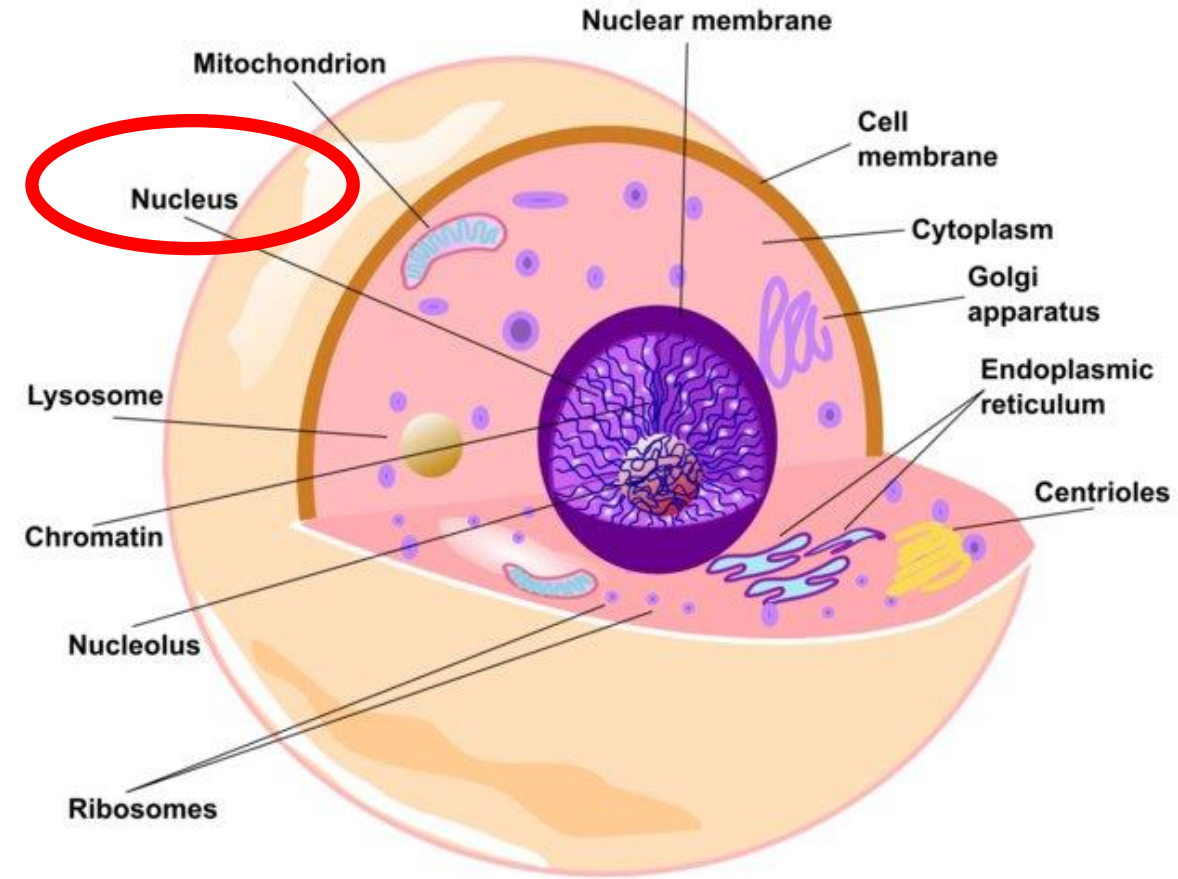


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A cell and its sub-cellular components.



Nucleus

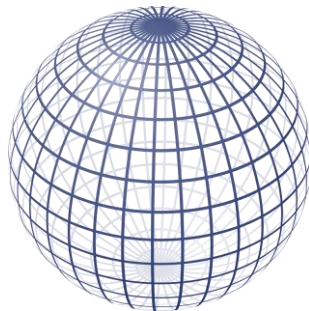
HeLa Cell lines

Nucleus

Discovery:

- **Robert Brown (1831) – discovered in the cells of orchids**
- Nucleus is a **darkly stainable, largest cell organelle** present in eukaryotic cells.
- It is usually **spherical**. It may be **lobed** in WBC; **kidney** shaped in paramecium.
- Nucleus has an outer double layered **nuclear membrane** with **nuclear pores**,
- Transparent granular matrix called **nucleoplasm**, **chromatin network** composed of DNA and histones and a darkly stainable spherical body called **Nucleolus**.

Spherical



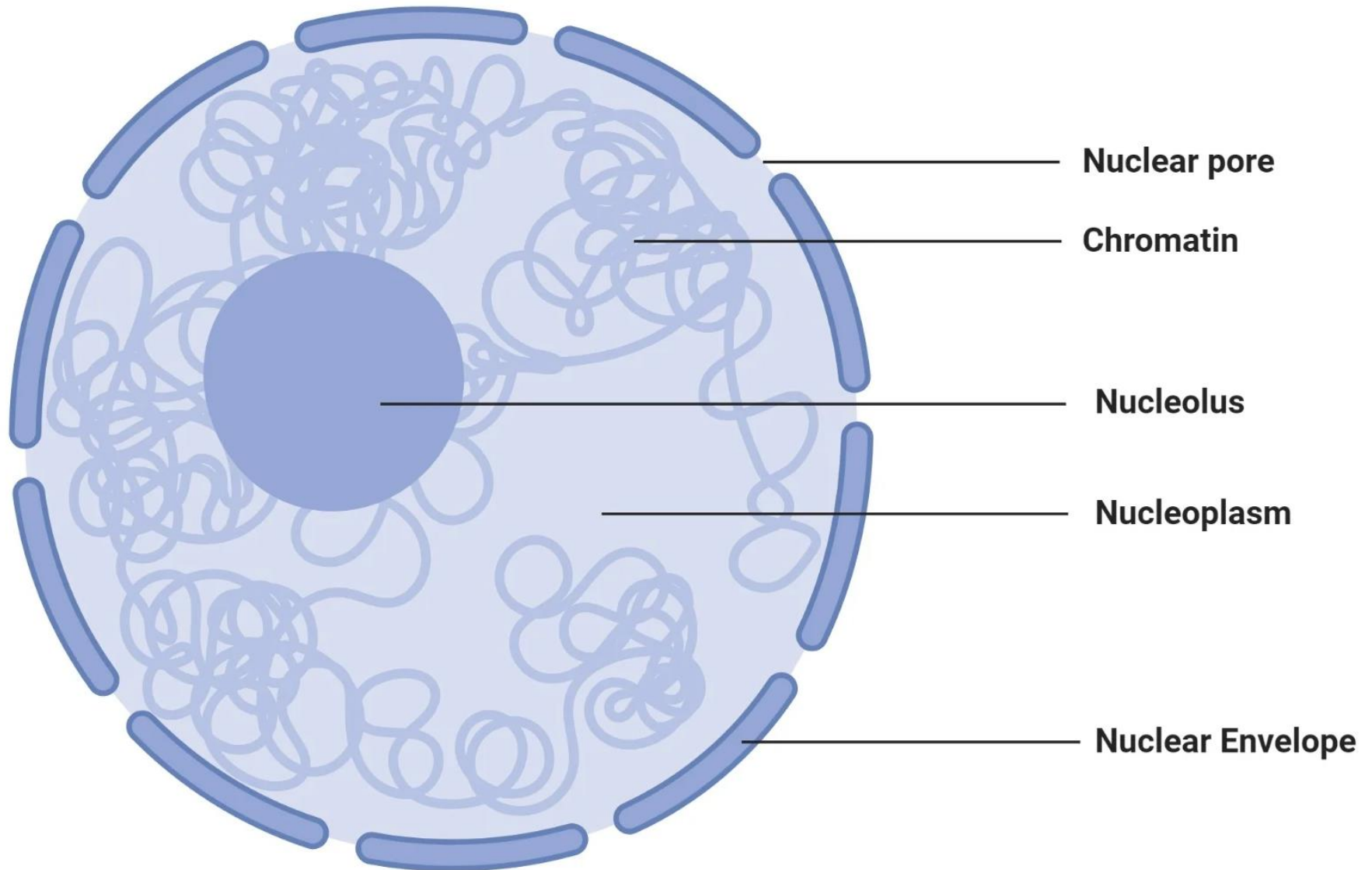
Lobular



Kidney



Nucleus and its components



Distribution of Nucleus

- **Enucleate cells:**

Cells without a nucleus are called enucleate cells. However, some living cells like mature sieve tubes of phloem and RBC's of mature mammals lack nuclei.

- **Uninucleate cell:**

It is also referred to as monokaryotic cells, mostly plant cells which contain a single nucleus.

- **Bi-nucleate cell:**

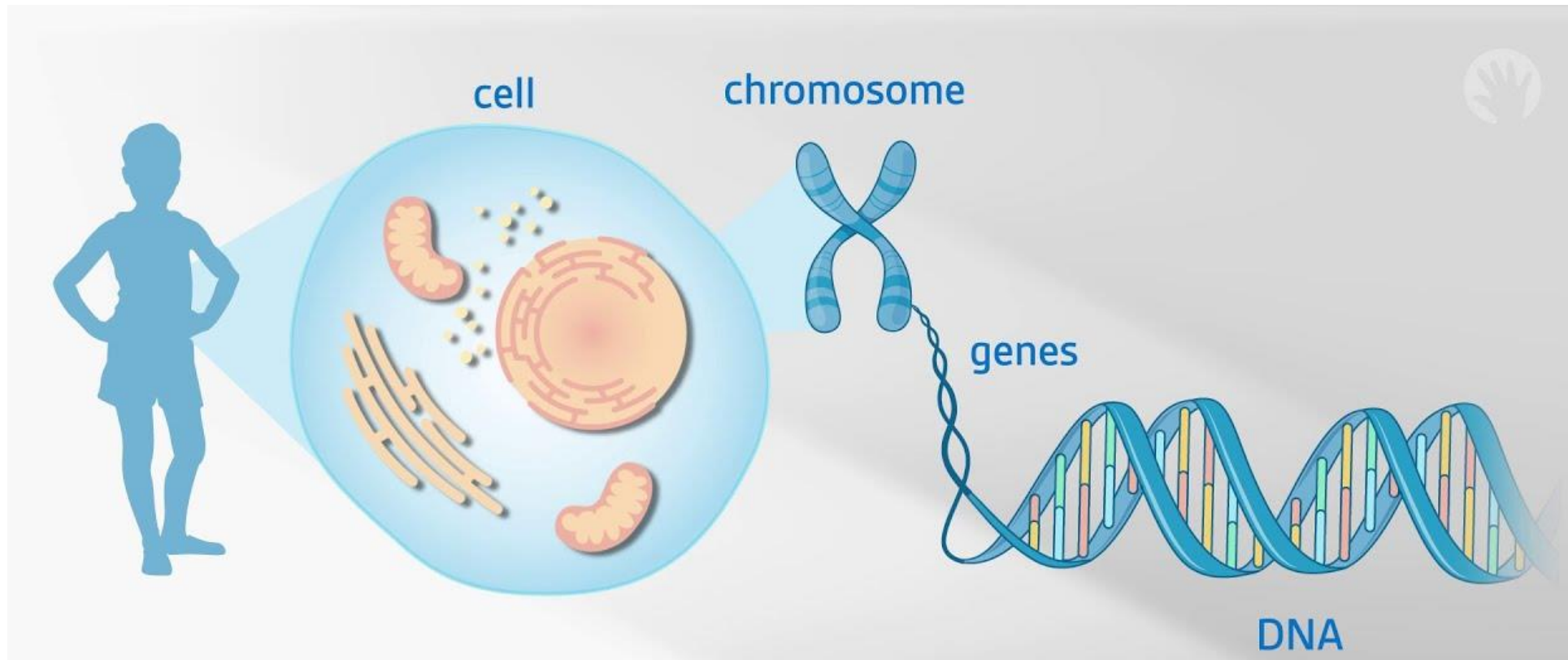
It is also called a dikaryotic cell. It contains two nuclei at a time. The examples are one paramecium (have mega and micronucleus), liver cells and cartilage cells.

- **Multinucleate cells:**

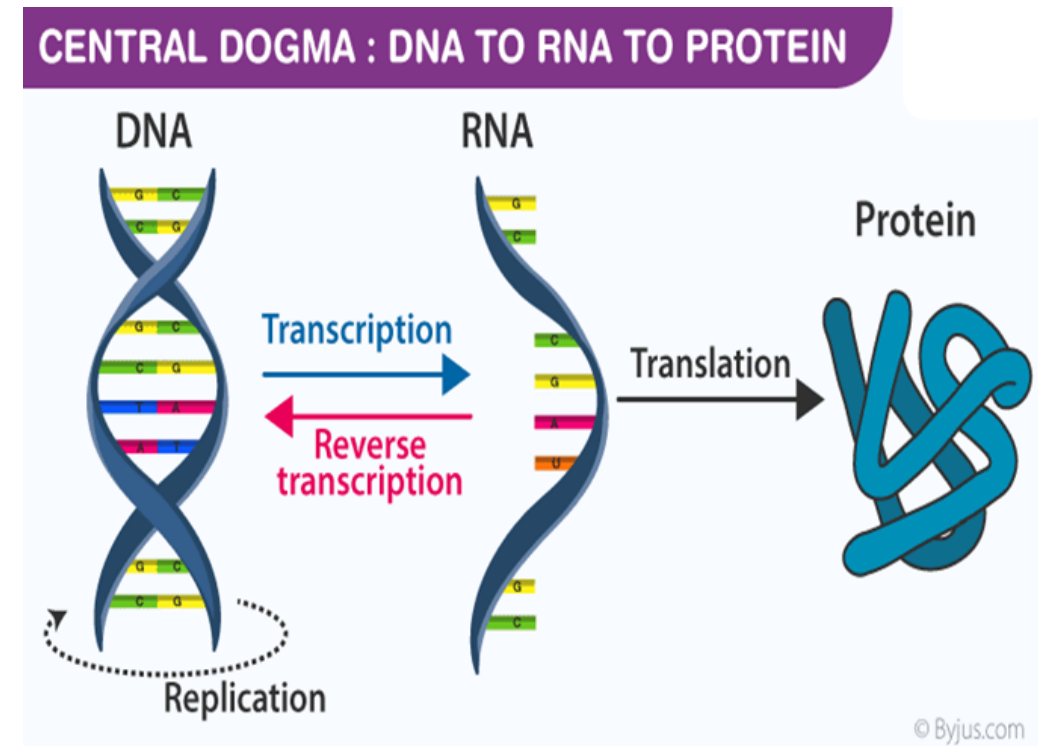
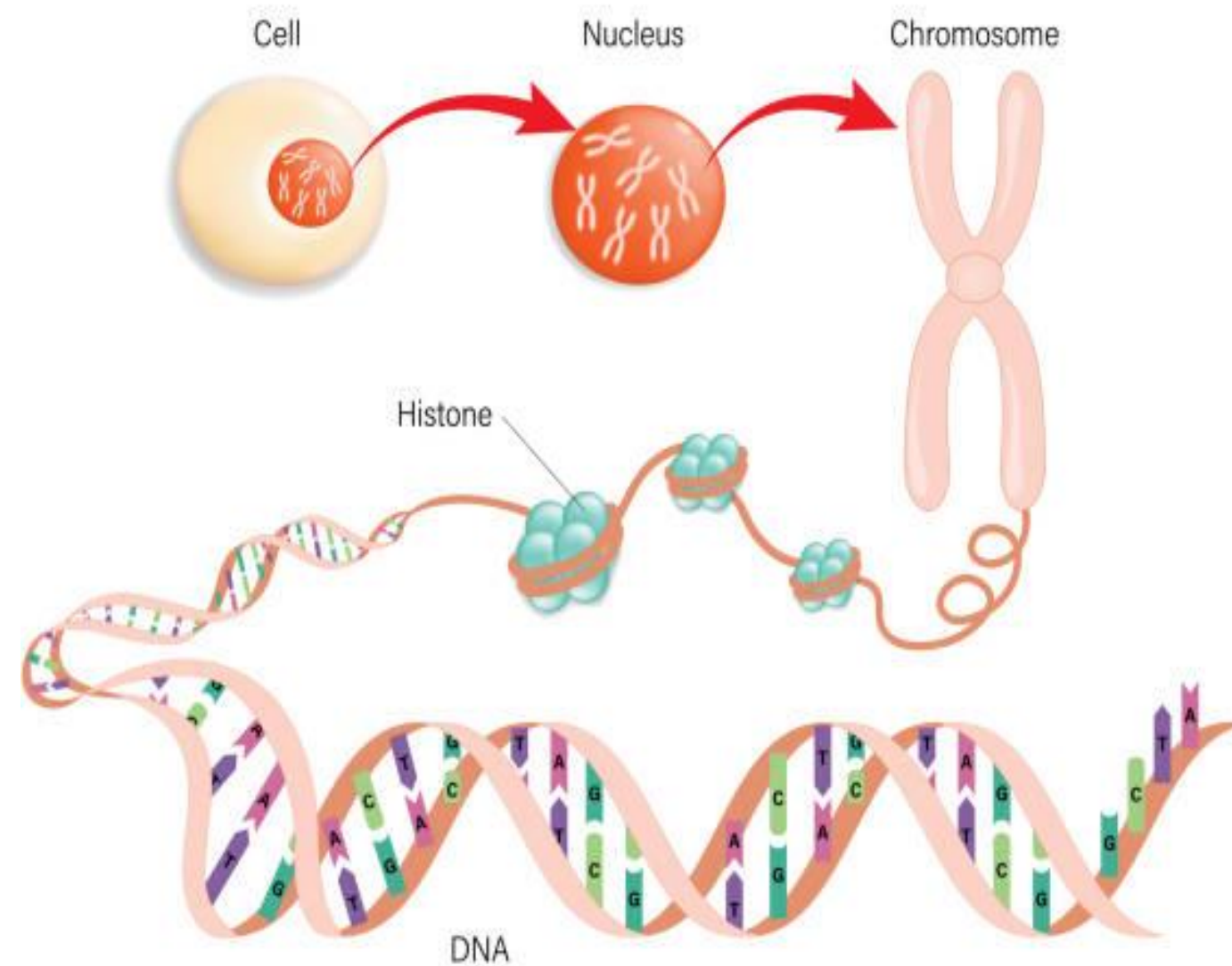
It is also known as the polynucleated cell which contains more than 2 nuclei at a time. For example, plants latex cells and latex vessels. In animals, striated muscle and bone marrow cells.

Cell components: **Nucleus**

- Nucleus is the **controlling centre of the cell**
- It contains the **genetic material DNA** which regulates various metabolic activities of the body by directing the synthesis of structural and functional proteins

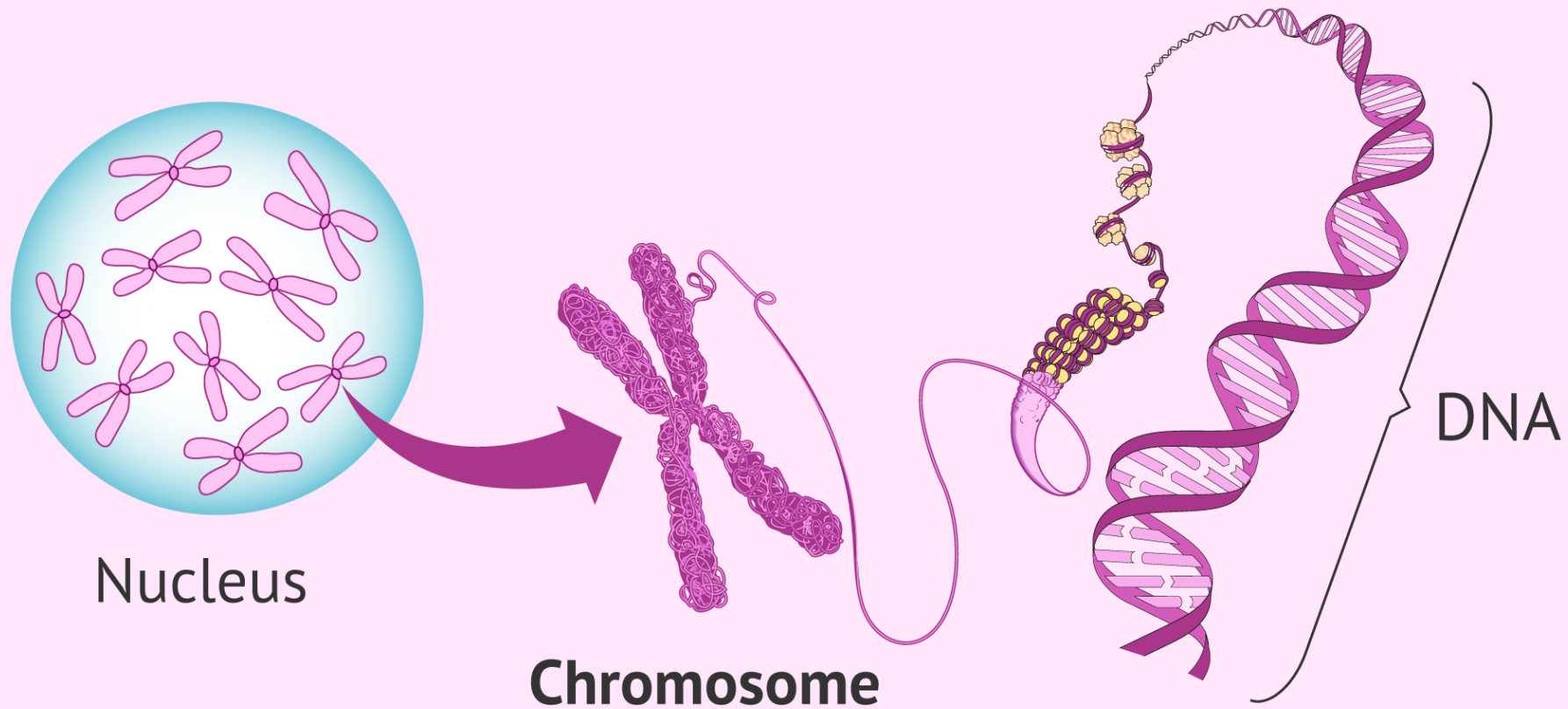


Cell, Nucleus, Chromosome, Histone and DNA

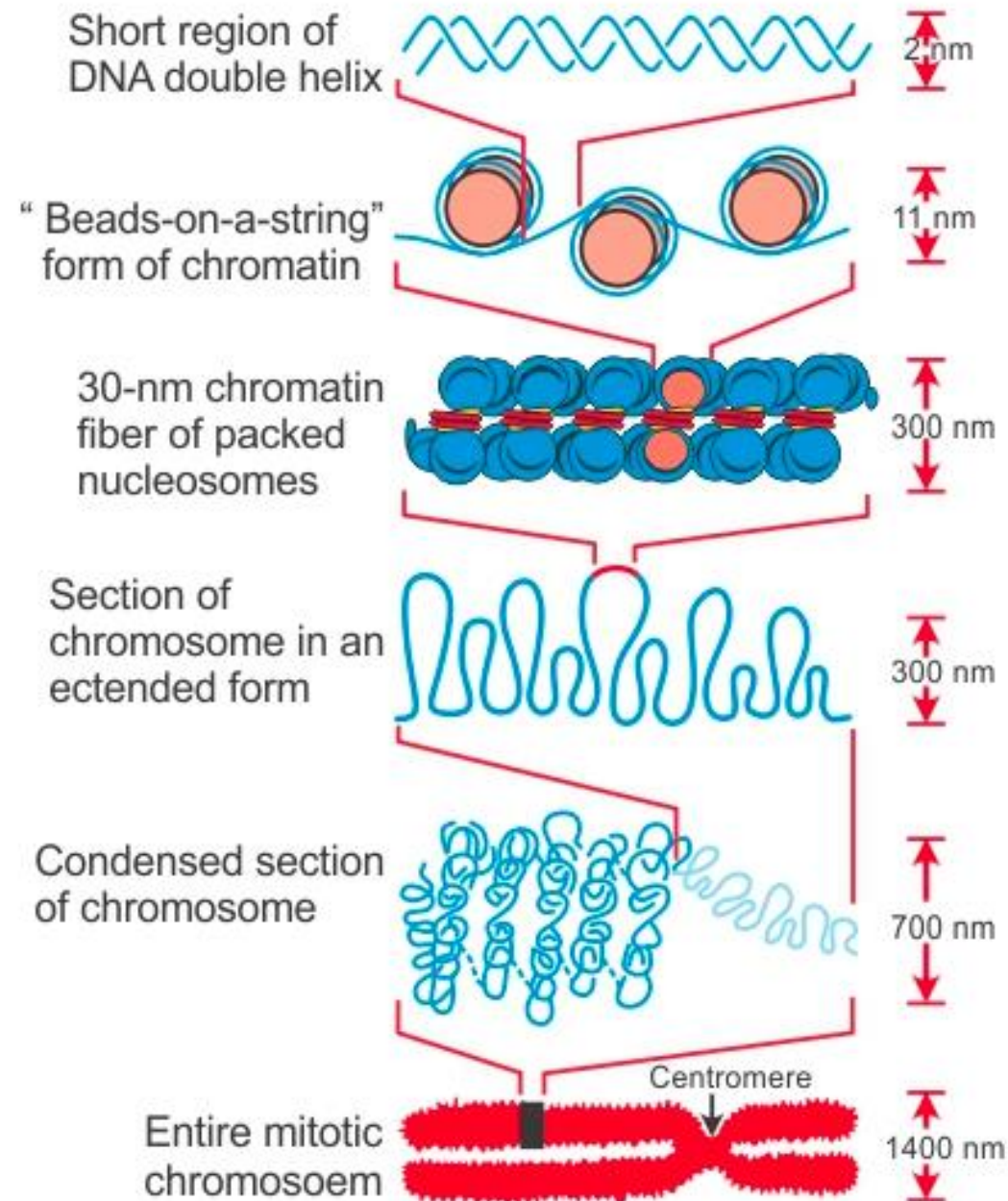


Cell components: **Chromosome**

- The nucleus of a normal or non dividing cell has a loosened indistinct network of nucleoprotein fibers called chromatin (coined by **Flemming**).
- During cell division the chromatin condenses to form distinctly visible chromosomes.



Cell components: Chromosome



Nuclear membrane, Nucleoplasm and Nucleolus

Nuclear Membrane:

The nuclear membrane is a **double-layered** membrane made up of lipoproteins that encloses the elements of the nucleus.

A **liquid-filled** space or **perinuclear** space is present between the two layers of a nuclear membrane.

The nucleus gets through the remaining of the cell or the cytoplasm through several openings called nuclear pores. Such nuclear pores are the sites for the **exchange of large molecules between the nucleus and cytoplasm**.

Nucleoplasm:

The nuclear membrane encloses the **clear, homogeneous, transparent** liquid.

Functions of nucleoplasm is to **maintain nuclear shape and structure**, and the **transportation** of *ions, molecules, nutrients*, and other substances important to cellular function.

Nuclear membrane, Nucleoplasm and Nucleolus

Nucleolus:

The nucleolus is a solid, spherical-shaped structure found inside/center of the nucleus. Some of the eukaryotic organisms have a nucleus that consists of up to four nucleoli.

The nucleolus is considered as the brain of the nucleus, covering nearly 25% volume of the nucleus.

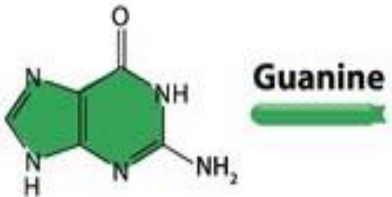
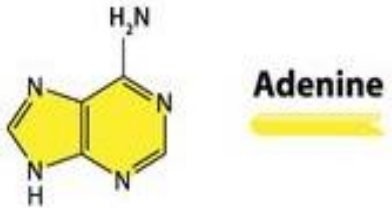
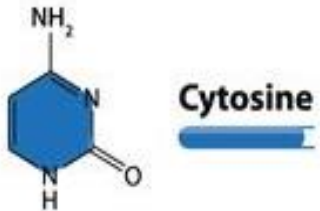
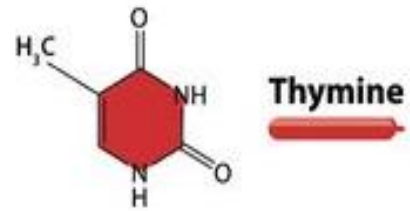
Primarily, it takes part in the **production of subunits that unites to form ribosomes.** Nucleolus plays an important role in the synthesis of proteins and in the production of ribosomes in eukaryotic cells.

Chromosome:

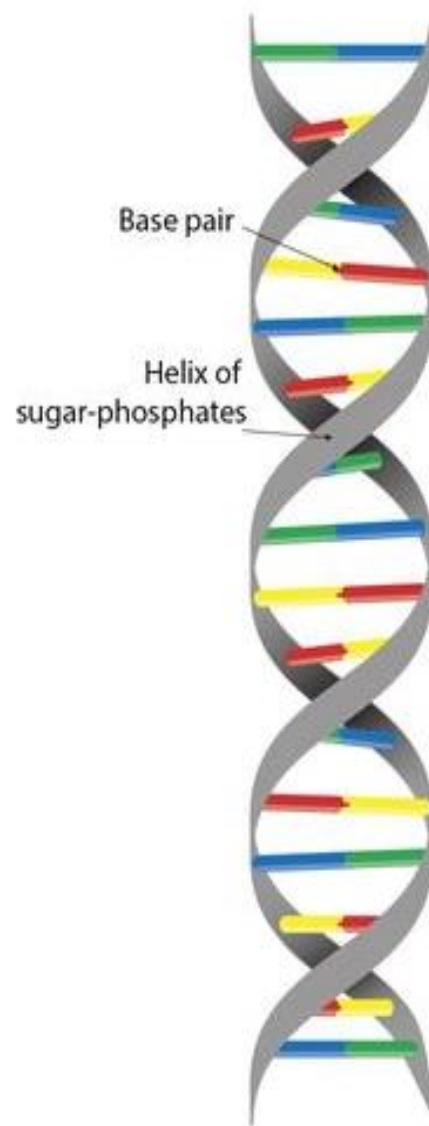
Chromosomes carry the **basic genetic material DNA** which is responsible to provide hereditary characteristics and genetic information to the various cells. The cellular functions are important for the growth and survival of living organisms.

Talking about the chemical composition of chromosomes, they are covered by histones and other proteins whose role is protection from chemical enzymes and physical forces. This, in turn, helps DNA to have a safeguard during the process of cell division.

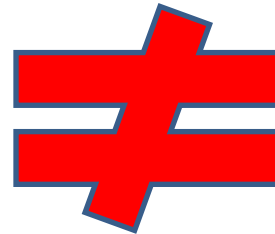
DNA, RNA and the Genetic Code



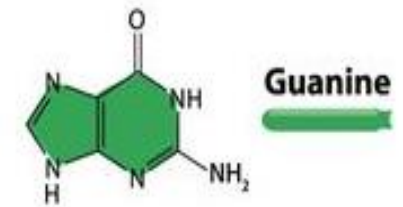
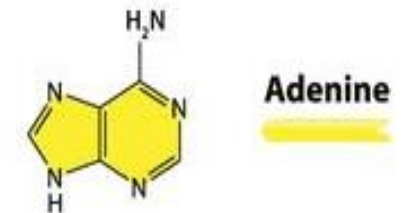
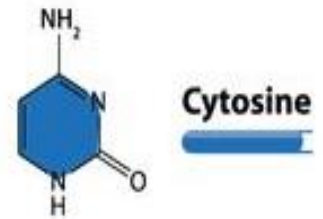
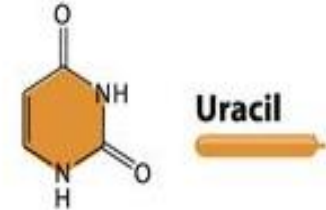
Nucleobases
of DNA



DNA
Deoxyribonucleic acid



RNA
Ribonucleic Acid



Nucleobases
of RNA

Summary<>Functions of the Nucleus

- 1) Nucleus is responsible for protein synthesis, growth, cell division, and differentiation.
- 2) The Nucleus is a site for transcription in which messenger RNA is produced for protein synthesis.
- 3) The nucleus functions as the exchange of hereditary molecules that is RNA and DNA between the nucleus and the rest of the cell.
- 4) Ribosome produced in the nucleolus.
- 5) The nucleus functions the selective transportation of regulatory factors and energy molecules through nuclear pores.
- 6) Chromatin is referred to as the storage of hereditary material, the genes in the form of DNA strands.
- 7) The Nucleolus is referred to as the storage of proteins and RNA in the nucleolus.
- 8) During cell division, chromatins are arranged into chromosomes in the nucleus.

Ribosome

Discovery: K R Porter (1945) - observed in animal cells.

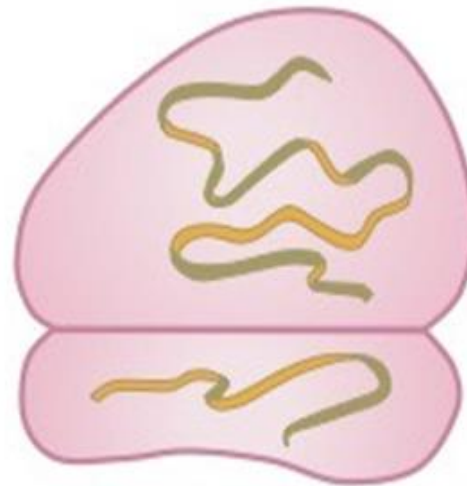
Robinson and Brown (1953) observed in plant cells

These are granular, **non-membranous sub spherical** structures present in the cytoplasm, mitochondria and chloroplast. They are also found attached to Rough ER and nuclear membrane.

Types of RNA



Messenger RNA (mRNA)



Ribosomal RNA (rRNA)



Transfer RNA (tRNA)

Function of Ribosomes

Ribosomes are the main site of polypeptide and protein synthesis

- The main function of ribosomes is to translate the messenger RNA (mRNA) encoded genetic information into proteins. Ribosomes move along a molecule of mRNA during protein synthesis, reading one codon at a time.
- The ribosome reading of each codon results in the incorporation of one amino acid into a progressively longer protein chain. The transfer of RNA (tRNA) molecules, which are the adapter molecules in the translation mechanism, brings the amino acids to the ribosome.
- Each of the 20 protein synthesis amino acids is connected to a specific type of tRNA. Thus, ribosomes can recognize and bind the right word for nucleic acid specified by the tRNA anticodon with its attached amino acid.
- Once the ribosome finally reaches a stop codon on the mRNA, the translation stops, separating and detaching the ribosomal subunits from the mRNA and releasing the completed protein.

Each amino acid, aside from its name, has a three-letter abbreviation and a one letter code.

Amino Acid	3 letter code	1 letter code	Amino Acid	3 letter code	1 letter code
Glycine	Gly	G	Threonine	Thr	T
Alanine	Ala	A	Cysteine	Cys	C
Valine	Val	V	Tyrosine	Tyr	Y
Leucine	Leu	L	Asparagine	Asn	N
Isoleucine	Ile	I	Glutamine	Gln	Q
Methionine	Met	M	Aspartic Acid	Asp	D
Proline	Pro	P	Glutamic Acid	Glu	E
Phenyl alanine	Phe	F	Lysine	Lys	K
Tryptophan	Trp	W	Arginine	Arg	R
Serine	Ser	S	Histidine	His	H

Proteins

A Polypeptide

Protein

Phosphofructokinase

Catalase

Fibrinogen

Serum albumin

Hemoglobin

Ovalbumin

Mass

345,400

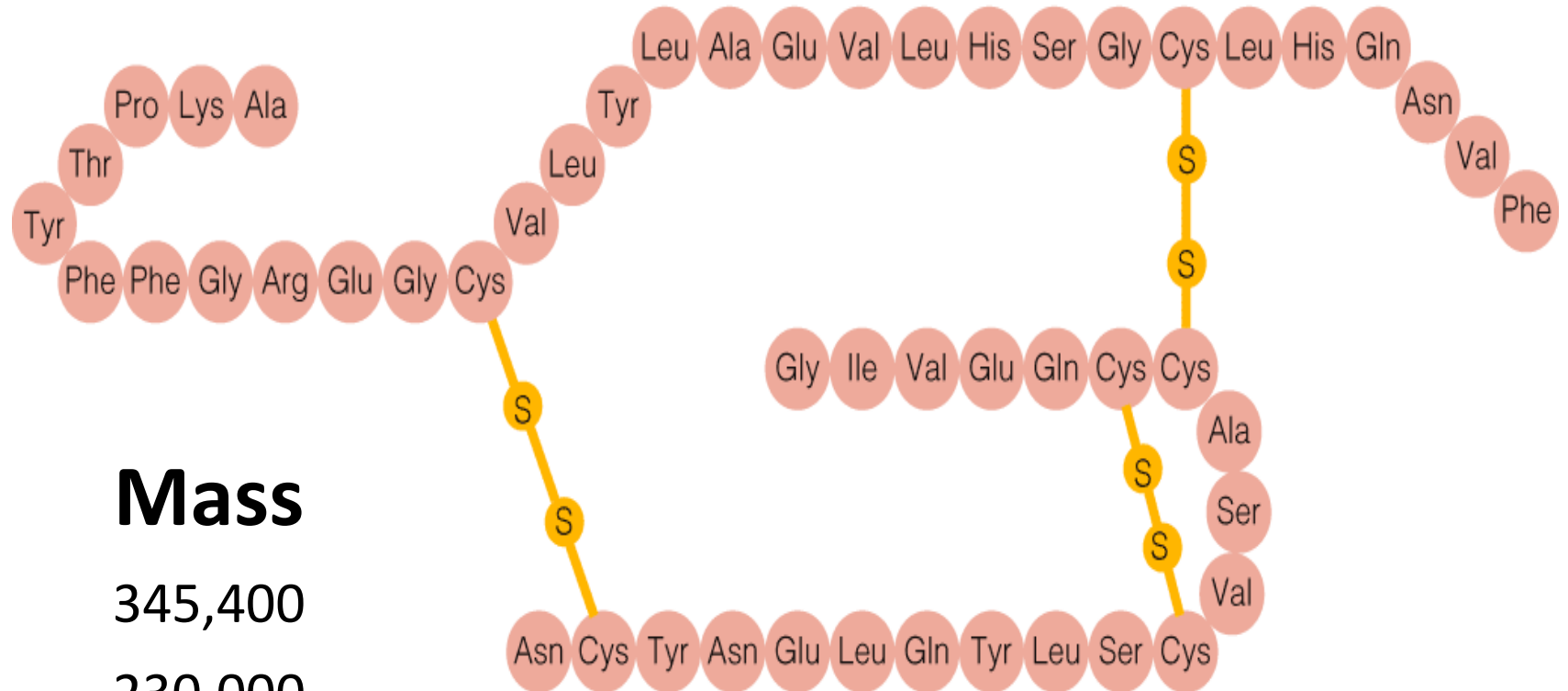
230,000

390,000

66,400

64,000

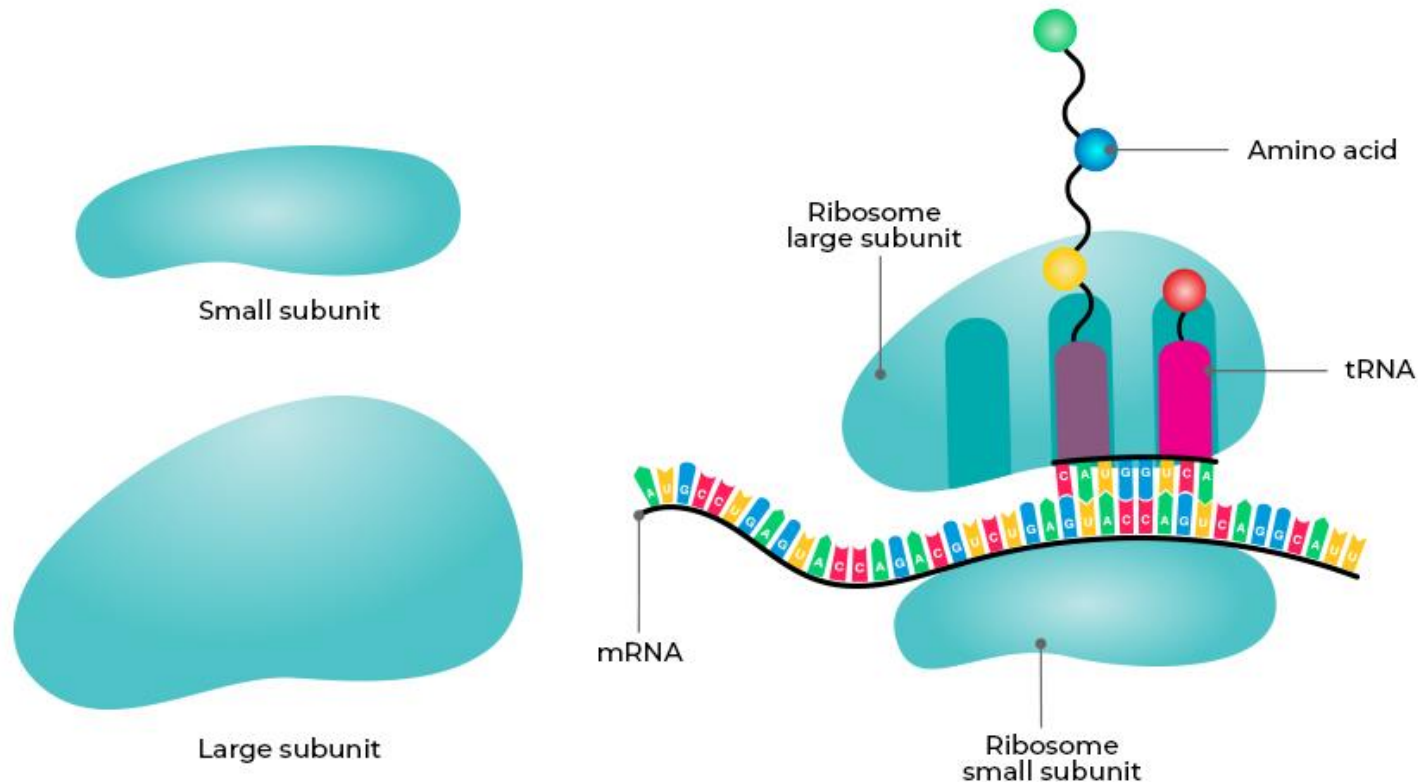
43,000



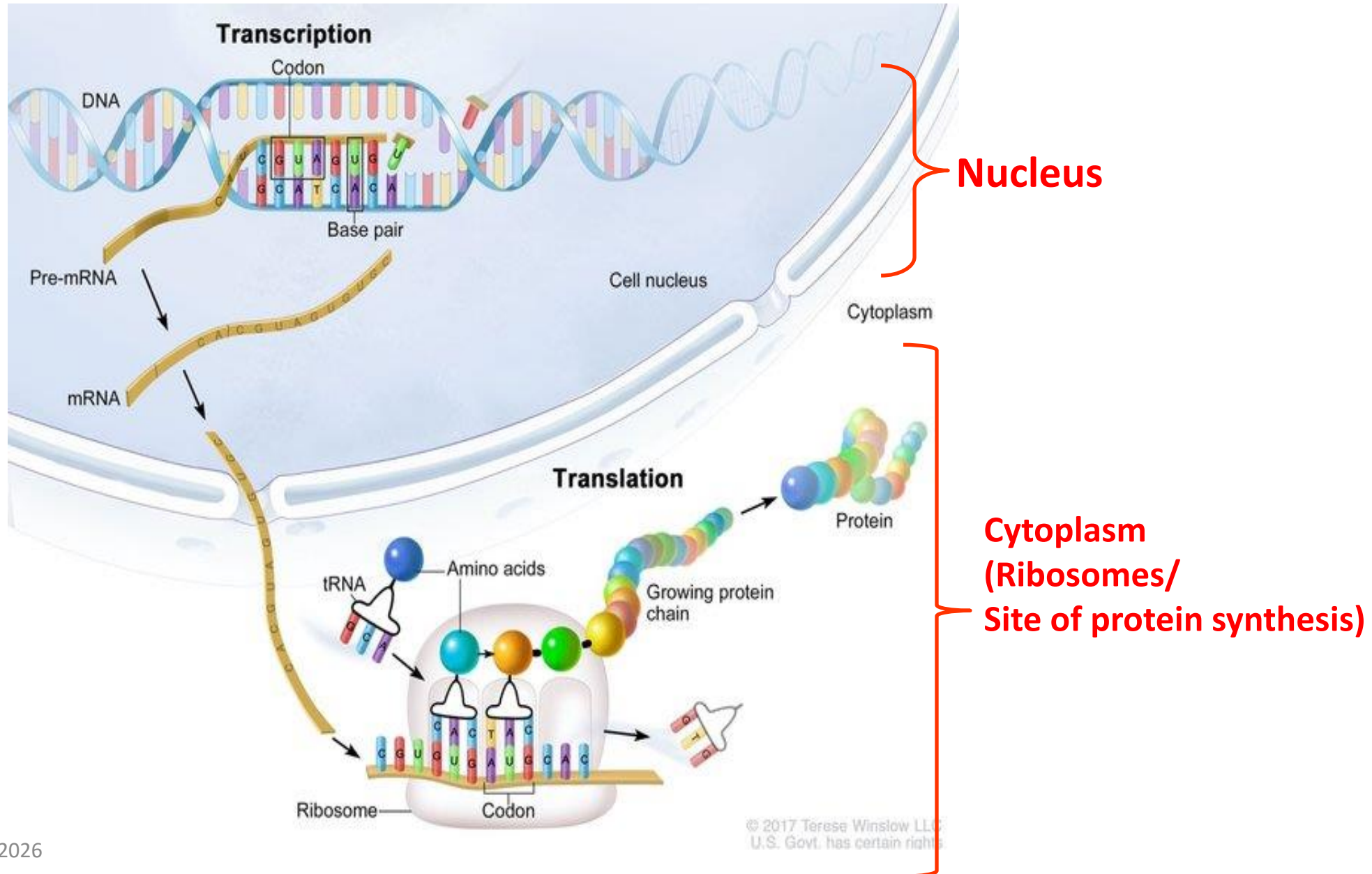
Cell components: Ribosome

Ribosome is a non membranous cell organelles. These are granular, non-membranous sub spherical structures present in the cytoplasm, mitochondria and chloroplast. They are also found attached to Rough ER and nuclear membrane.

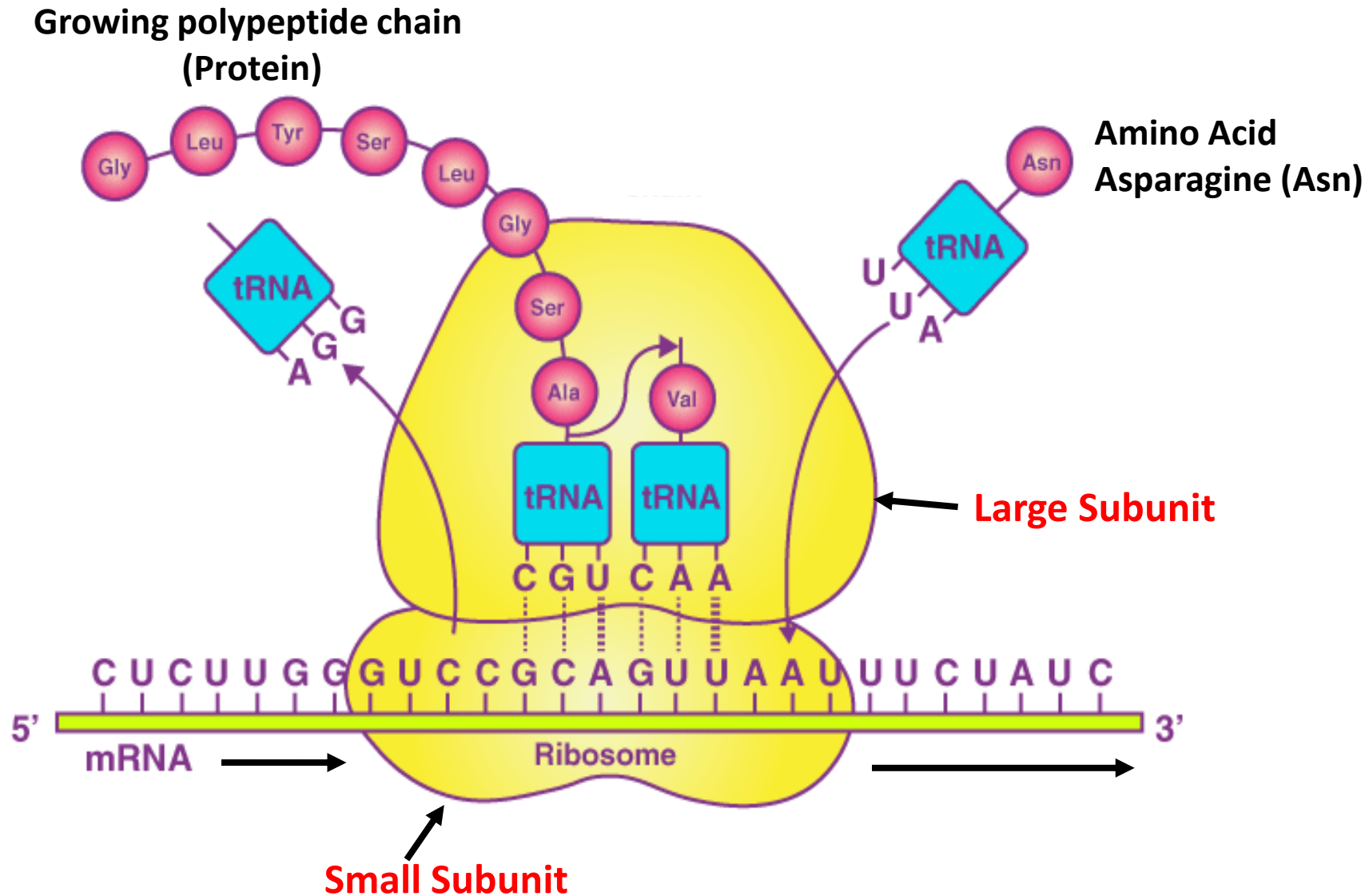
Function: These are the sites of polypeptide or protein synthesis



Mechanism of gene transcription and translation



How Protein Synthesis in Ribosomes?



Supplementary reading

Movement Through the Membrane

- Materials can move through the membrane by:

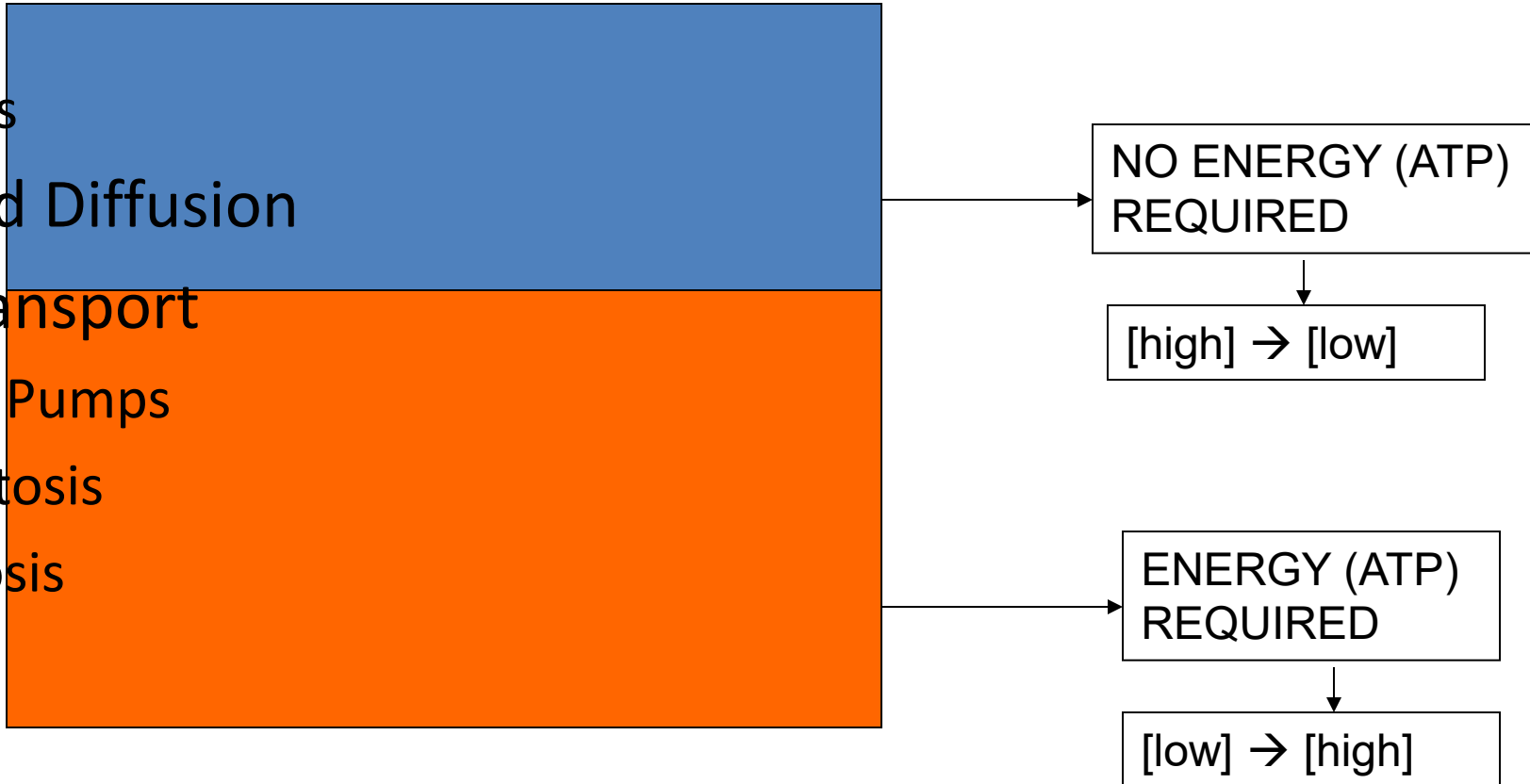
- Diffusion

- Osmosis

- Facilitated Diffusion

- Active Transport

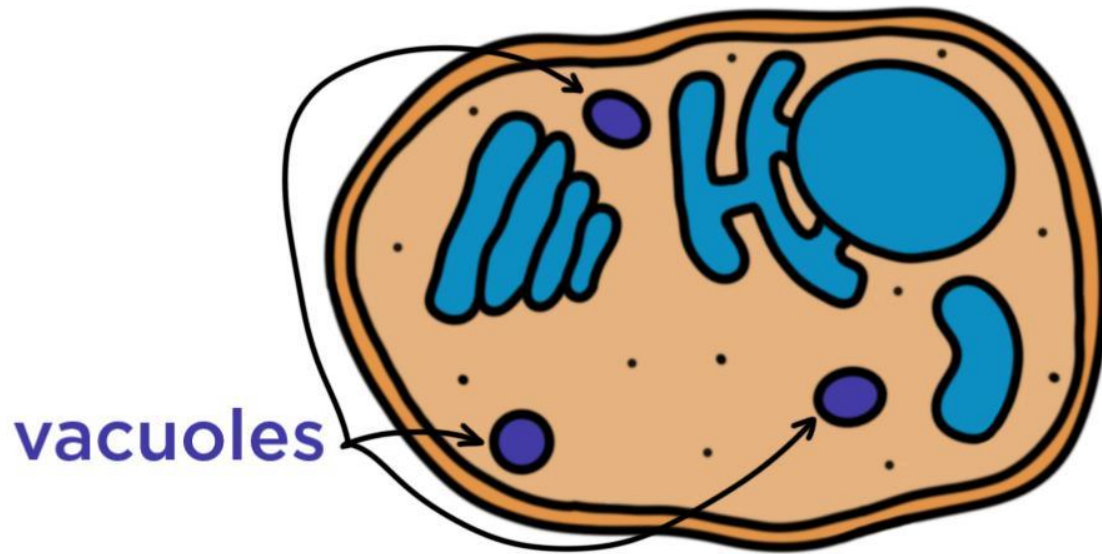
- Protein Pumps
 - Endocytosis
 - Exocytosis



Cell components: Vacuoles

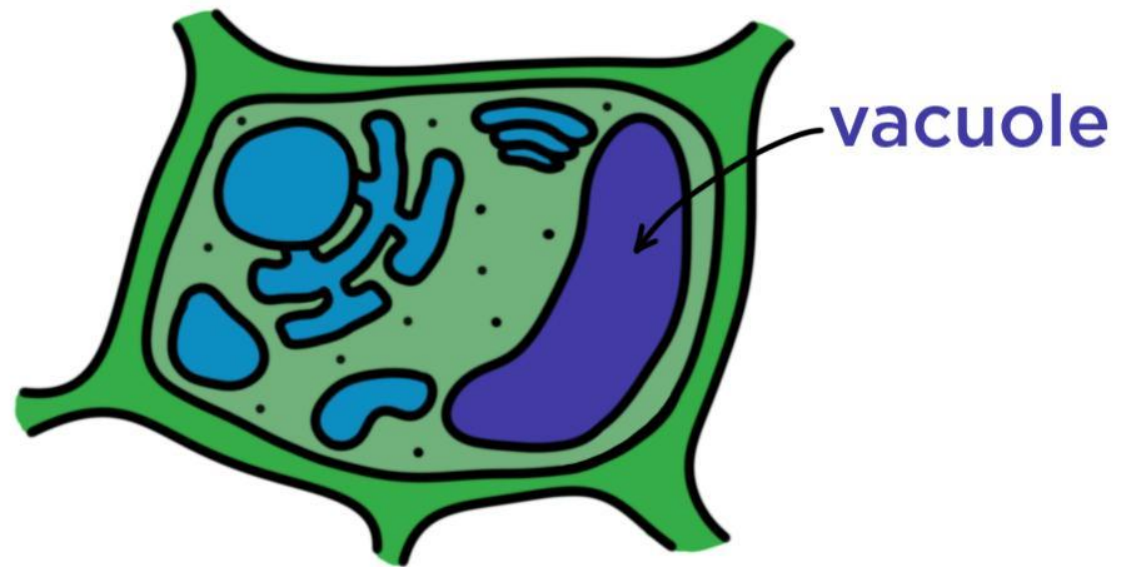
ORGANELLES: VACUOLE

ANIMAL CELL



- several small vacuoles
- used for storage
- can contain nutrients, water, or waste

PLANT CELL

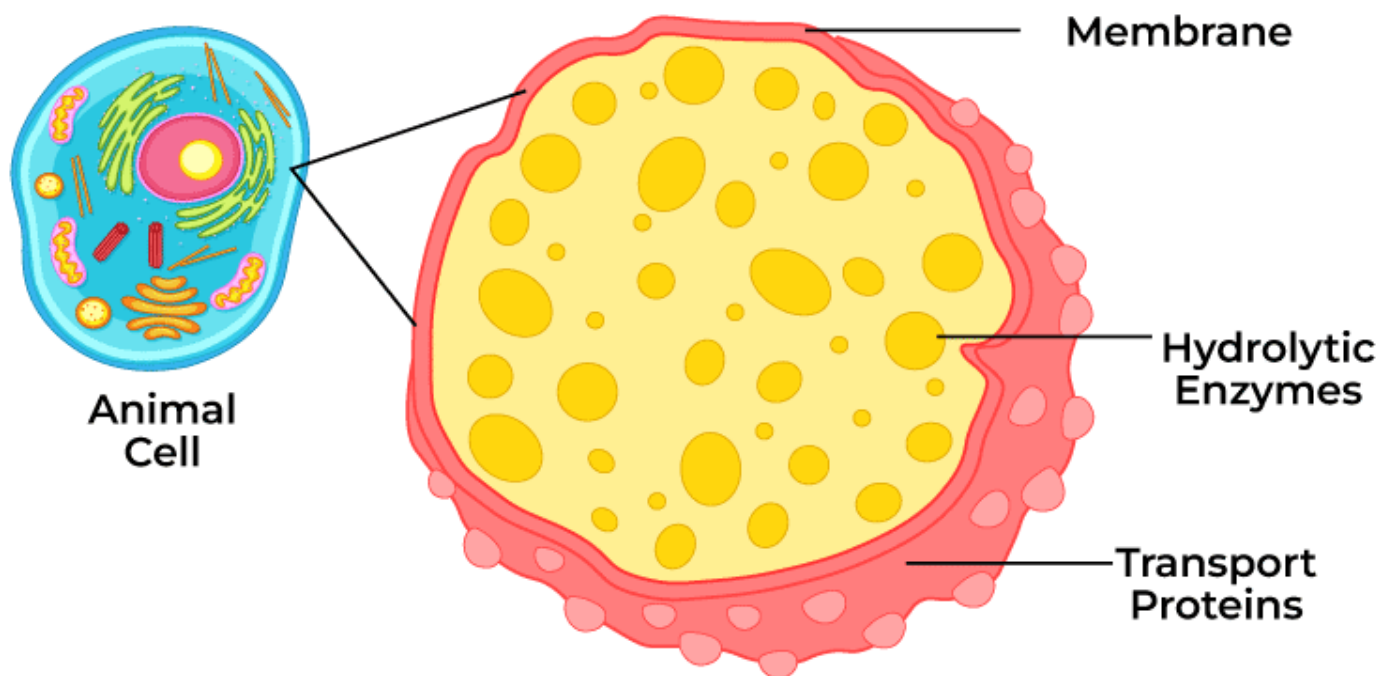


- one large vacuole
- used to store water and push against the cell wall
- keeps the plant rigid

Cell components: **Microbodies**

These are small, spherical, single membrane bound structures present in cytoplasm. The different types of microbodies are :

Lysosomes: Lysosomes are first reported by Belgian scientist **Christian de Duve** (1995) in rat liver cells. **Nivicott** (1950) coined Lysosomes. These are small single membrane bound vesicles filled with **hydrolytic enzymes**. Lysosomes are produced from Golgi complex .



Cell components: **Overview**

Cell Organelle and its Functions
Nucleolus
The nucleolus is the site of ribosome synthesis. Also, it is involved in controlling cellular activities and cellular reproduction
Nuclear membrane
The nuclear membrane protects the nucleus by forming a boundary between the nucleus and other cell organelles.

Cell components: Overview

Chromosomes

Chromosomes play a crucial role in determining the sex of an individual. Each human cells contain 23 pairs of chromosomes

Endoplasmic reticulum

The endoplasmic reticulum is involved in the transportation of substances throughout the cell. It plays a primary role in the metabolism of carbohydrates, synthesis of lipids, steroids and proteins.

Golgi Bodies

Golgi bodies are called the cell's post office as it is involved in the transportation of materials within the cell

Cell components: **Overview**

Ribosome

Ribosomes are the protein synthesisers of the cell

Mitochondria

The mitochondrion is called “the powerhouse of the cell.” It is called so because it produces ATP – the cell’s energy currency

Lysosomes

Lysosomes protect the cell by engulfing the foreign bodies entering the cell and helps in cell renewal. Therefore, it is known as the cell’s suicide bags

Cell components: **Overview**

Chloroplast

Chloroplasts are the primary organelles for photosynthesis. It contains the pigment chlorophyll

Vacuoles

Vacuoles stores food, water, and other waste materials in the cell

Cell: Function

Provides Support and Structure

- All the organisms are made up of cells. They form the structural basis of all the organisms. The cell wall and the cell membrane are the main components that function to provide support and structure to the organism. For eg., the skin is made up of a large number of cells. Xylem present in the vascular plants is made of cells that provide structural support to the plants.

Facilitate Growth Mitosis

- In the process of mitosis, the parent cell divides into the daughter cells. Thus, the cells multiply and facilitate the growth in an organism.

Cell : Function

Allows Transport of Substances

- Various nutrients are imported by the cells to carry out various chemical processes going on inside the cells. The waste produced by the chemical processes is eliminated from the cells by active and passive transport. Small molecules such as oxygen, carbon dioxide, and ethanol diffuse across the cell membrane along the concentration gradient. This is known as passive transport. The larger molecules diffuse across the cell membrane through active transport where the cells require a lot of energy to transport the substances.

Cell : Function

Energy Production

- Cells require energy to carry out various chemical processes. This energy is produced by the cells through a process called photosynthesis in plants and respiration in animals.

Aids in Reproduction

- A cell aids in reproduction through the processes called mitosis and meiosis. Mitosis is termed as the asexual reproduction where the parent cell divides to form daughter cells. Meiosis causes the daughter cells to be genetically different from the parent cells. Thus, we can understand why cells are known as the structural and functional unit of life. This is because they are responsible for providing structure to the organisms and performs several functions necessary for carrying out life's processes.

Cell Cycle Regulators

- **Cyclin**
 - Protein that regulates the cell cycle in eukaryotic cells
 - When injected into a non-dividing cell it causes a mitotic spindle to form
- **Internal Regulators**
 - Responds to events inside the cell
 - Makes sure that a cell does not enter mitosis until all chromosomes are replicated

Cell Cycle Regulators

- **External Regulators**
 - Respond to events outside the cell
 - “Growth factors” that speed up or slow down growth and division

Uncontrolled Cell Growth

- **Cancer:**
 - Cells that lose the ability to control cell growth
 - Most cancers have damage to the p53 gene
 - Normally halts the cell cycle until all chromosomes are replicated
 - Chromosome damage builds up and the cancer cell loses the information that controls normal cell growth
 - Tumors → masses of cells that can damage the surrounding tissue
 - **CAUSES:** smoking tobacco, radiation exposure (UV, XRAY, etc.), viral infection

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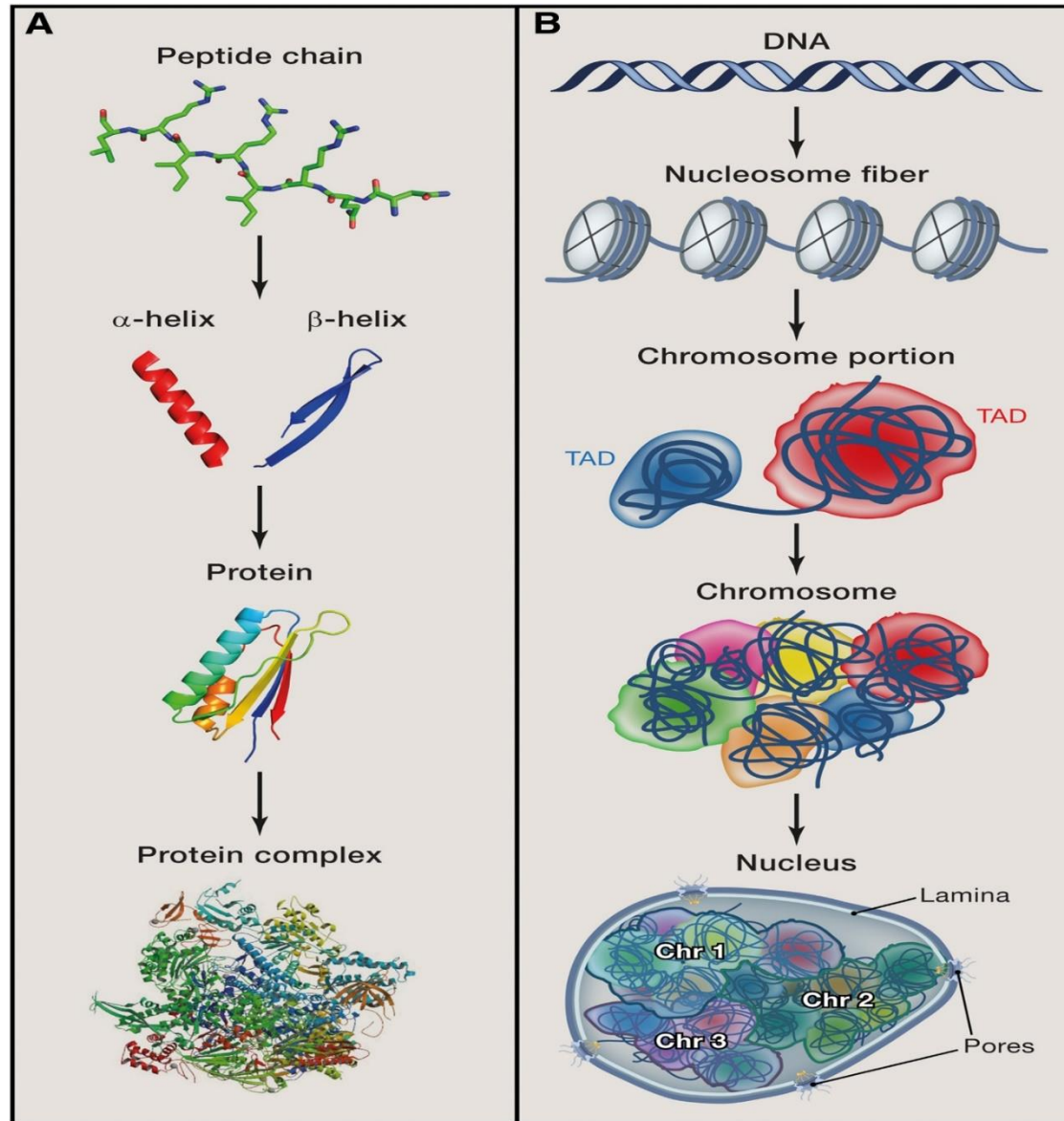
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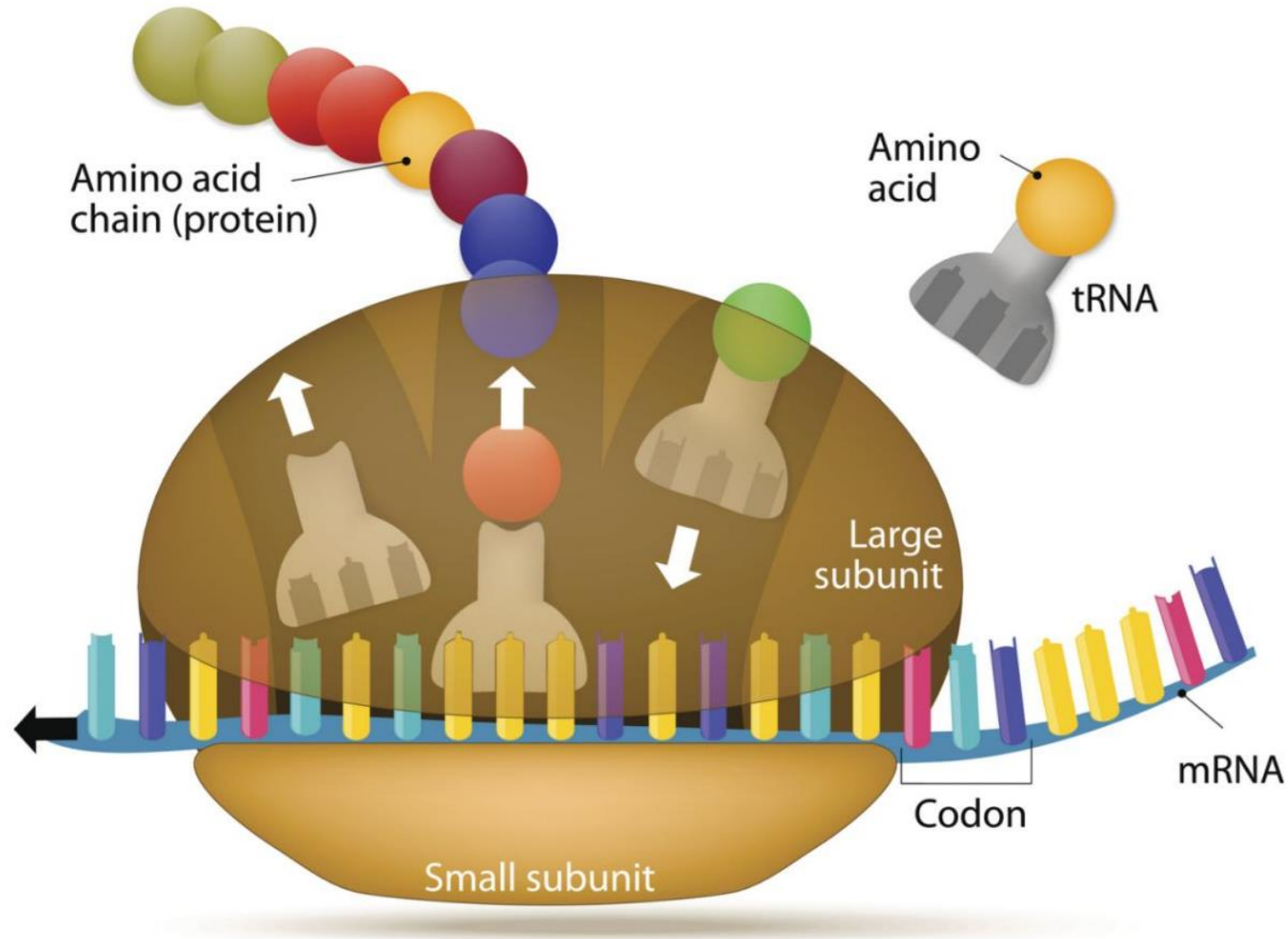
Characteristics of cells

- Cells provide the necessary structural support for an organism.
- The genetic information necessary for reproduction is present within the nucleus.
- Structurally, the cell has cell organelles which are suspended in the cytoplasm.
- Mitochondria is the organelle responsible for fulfilling the cell's energy requirements.
- Lysosomes digest metabolic wastes and foreign particles in the cell.
- Endoplasmic reticulum synthesises selective molecules and processes them, eventually directing them to their appropriate locations.

Chromosome



Protein synthesis in Ribosome



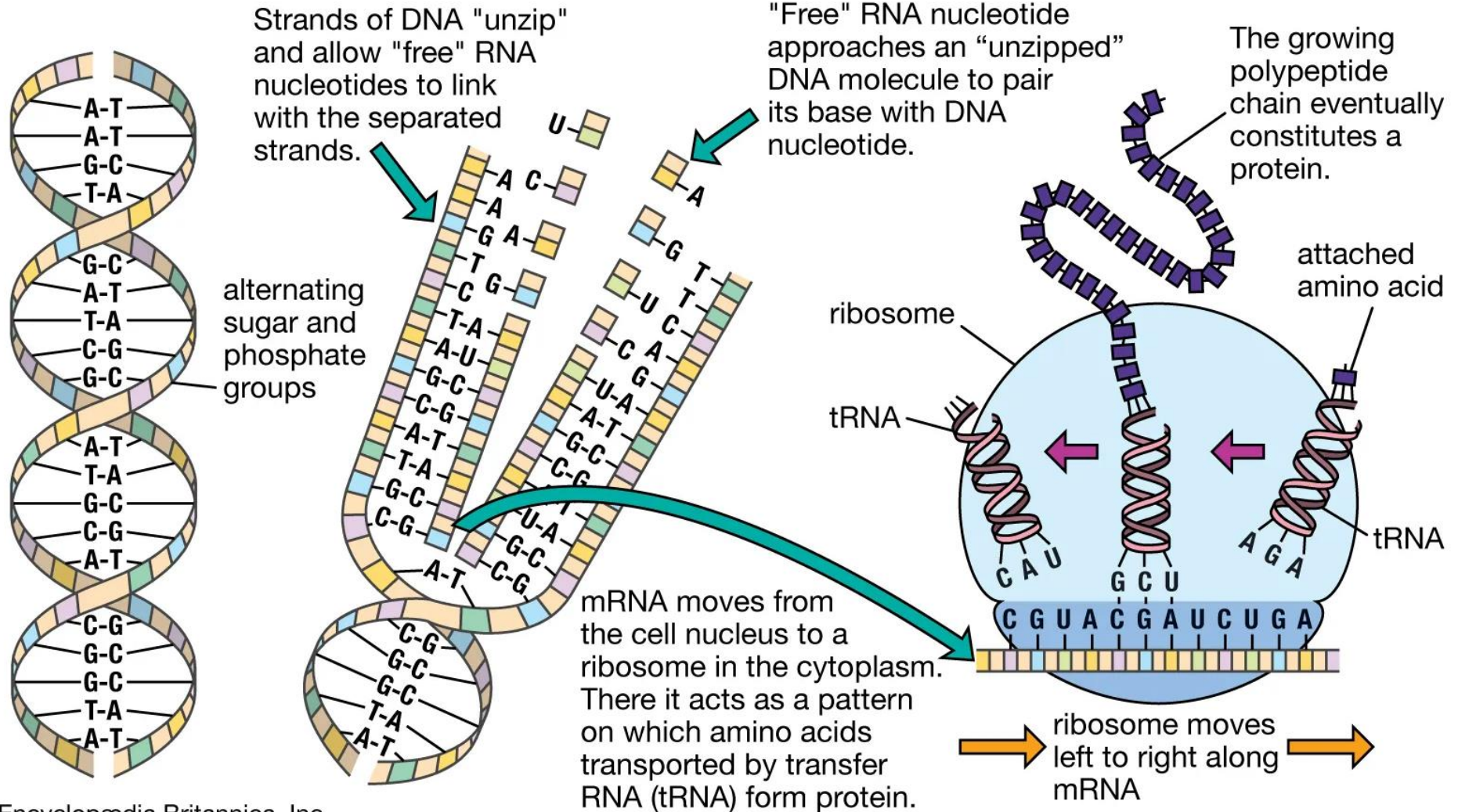
<https://www.youtube.com/watch?v=Ikq9AcBcohA>

How DNA directs protein Synthesis

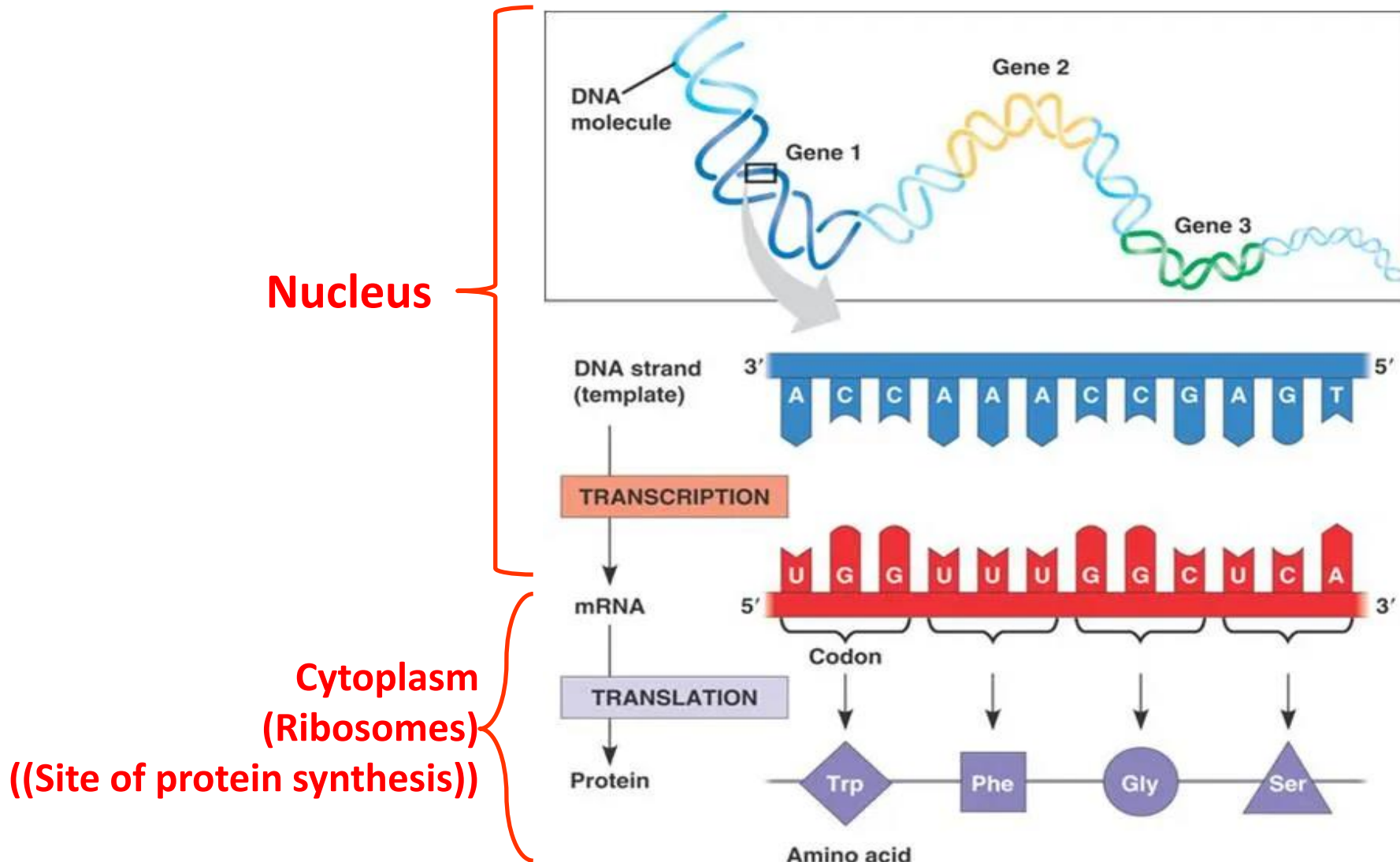
1. Double-stranded DNA in the cell nucleus

2. Messenger RNA (mRNA) forming on DNA strands

3. Formation of protein on ribosome

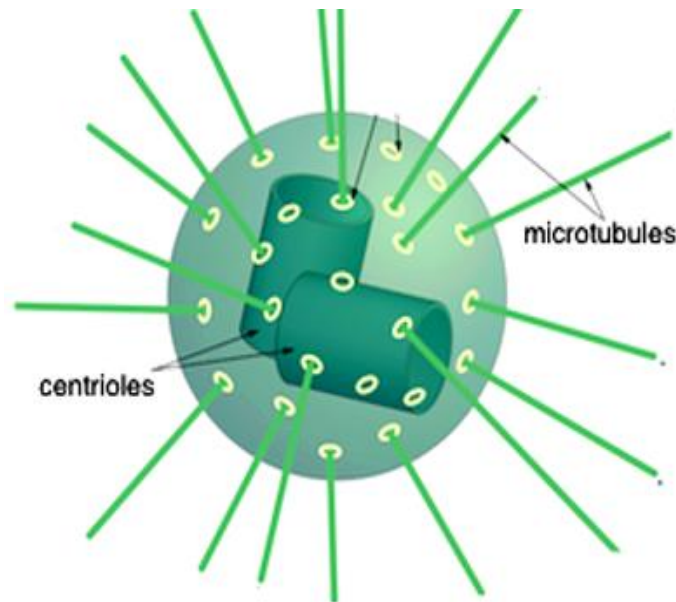


Transcription and Translation and Protein Production



Cell components: Centrosome

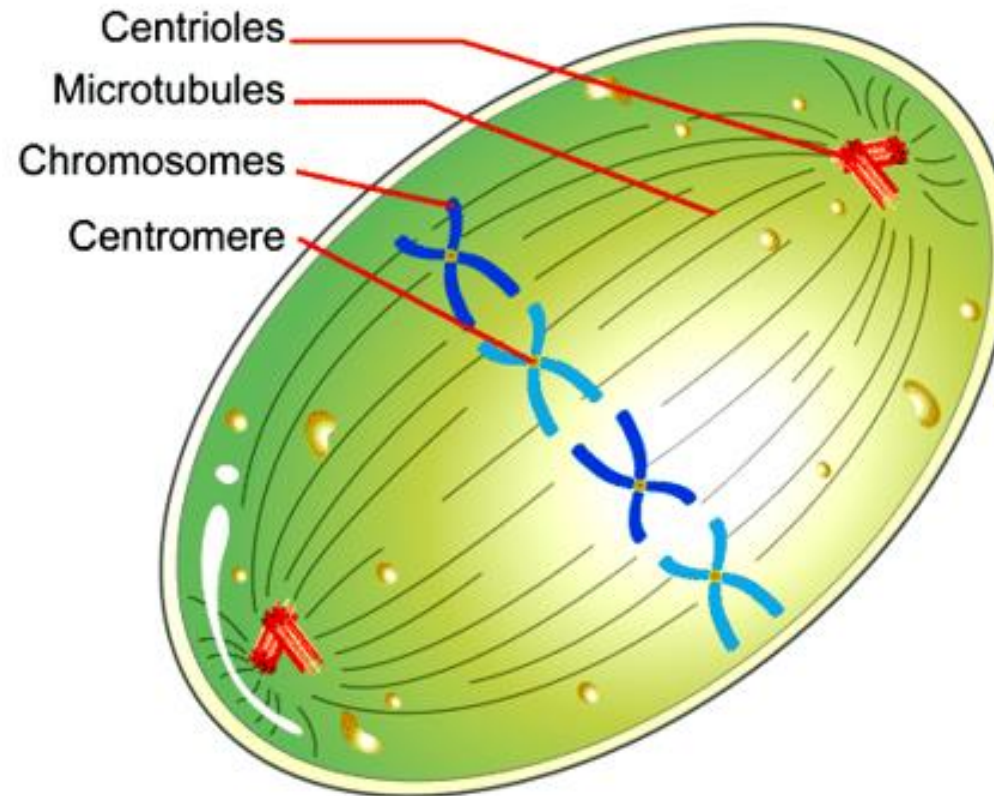
Centrosome is found in **animal cells** and in some motile algae. It is **absent in plant cells**. It is present near the nucleus. Centrosome has two cylindrical structures called **centrioles** surrounded by a less denser cytosol called **centrosphere**. The centrioles are arranged at right angles to one another. Each centriole is made up of a whorl of nine triplets of microtubules. These microtubules run parallel to one another. The adjacent microtubules are connected by proteinaceous strands.



Cell components: Centrosome

Functions:

- They form asters and organize the formation of spindle fibres during cell division.
- They are involved in the formation of cilia and flagella. Centrioles are present in sperms.



Cell components: **Non-living Inclusions**

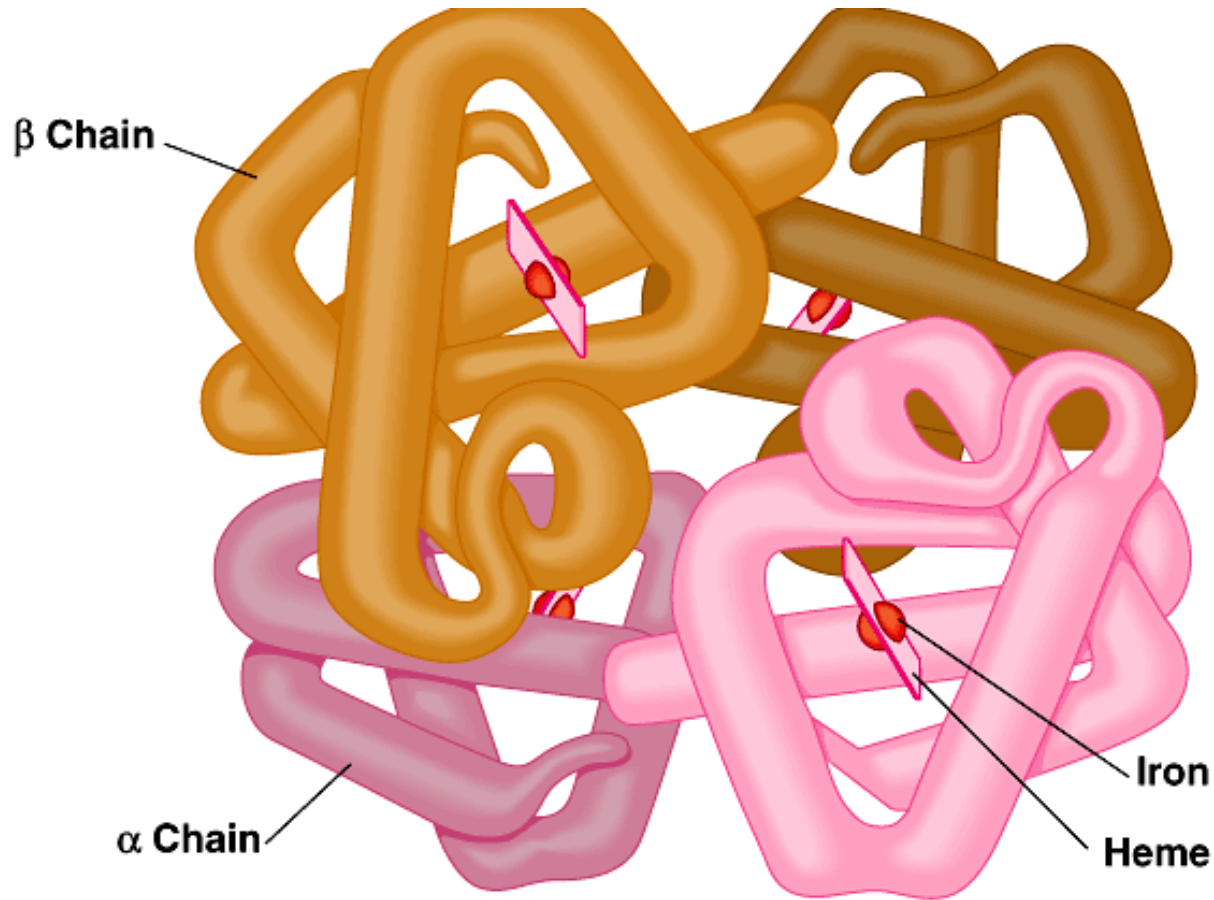
The non living cell inclusions includes **ergastic substances** and **cytoskeleton elements**

1. Ergastic substances: These are non living cell inclusions of cytoplasm like reserve food materials (starch, protein, oils), secretory products (nectar, pigments, enzymes), excretory products (alkaloids, resins, latex, tannins) and mineral crystals (cystoliths, raphides, druses).

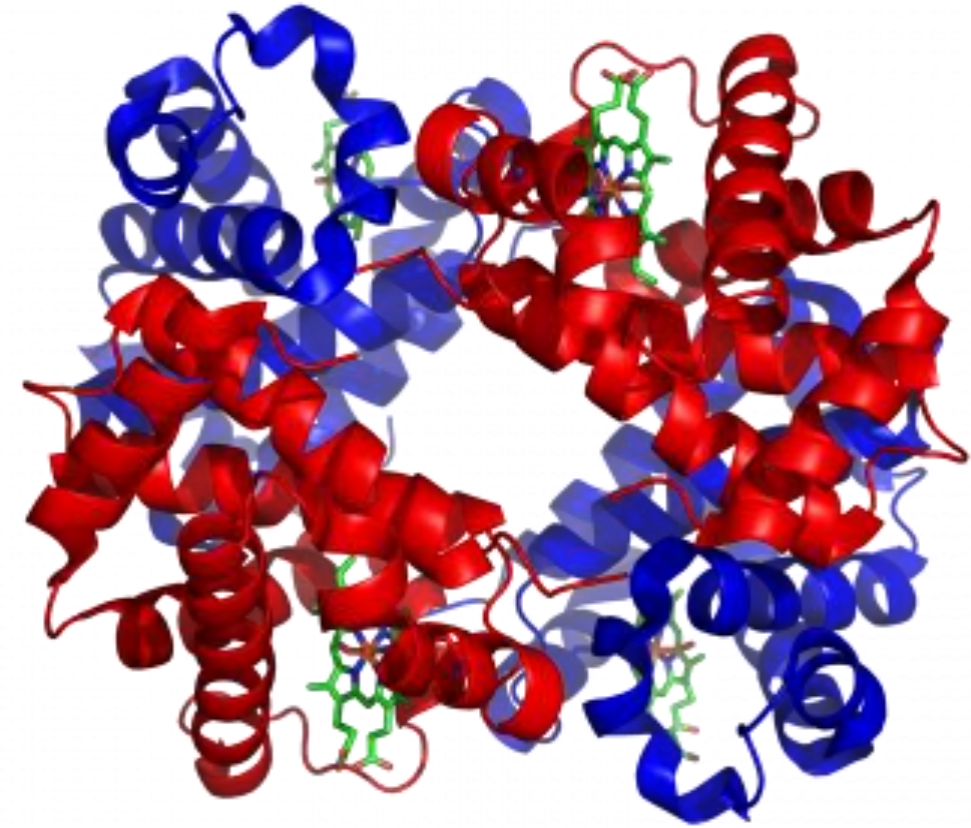
2. Cytoskeleton: It is a complex network of interconnected **microfilaments** and **microtubules** of protein fibres present in cytoplasm. The microfilaments are composed of **actin** and microtubules are composed of **tubulins**.

- It helps in mechanical support, cell motility, cell division and maintenance of the shape of the cell.

The quaternary structure of proteins



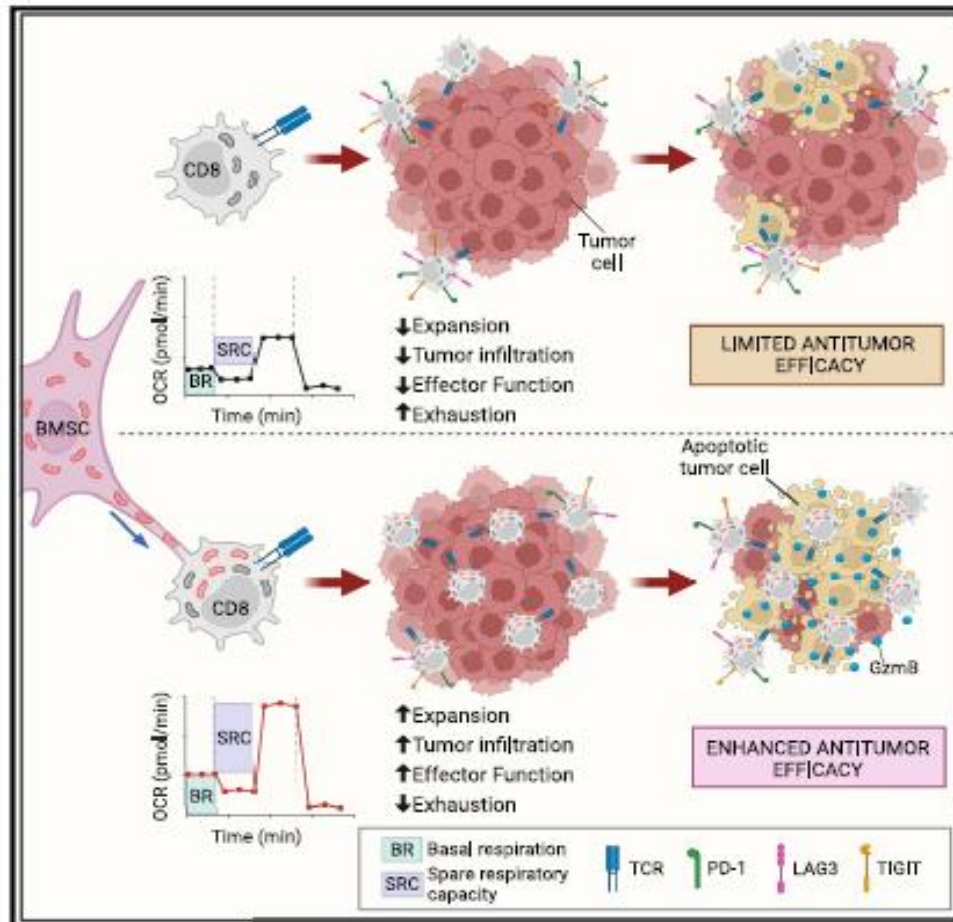
(b) Hemoglobin



Human Hemoglobin is composed of four polypeptides, two alpha subunits (shown in red) and two beta subunits (shown in blue).

Intercellular nanotube-mediated mitochondrial transfer enhances T cell metabolic fitness and antitumor efficacy

Graphical abstract



Authors

Jeremy G. Baldwin,
Christoph Heuser-Loy, Tanmoy Saha, ...,
Jessica Fioravanti, Shiladitya Sengupta,
Luca Gattinoni

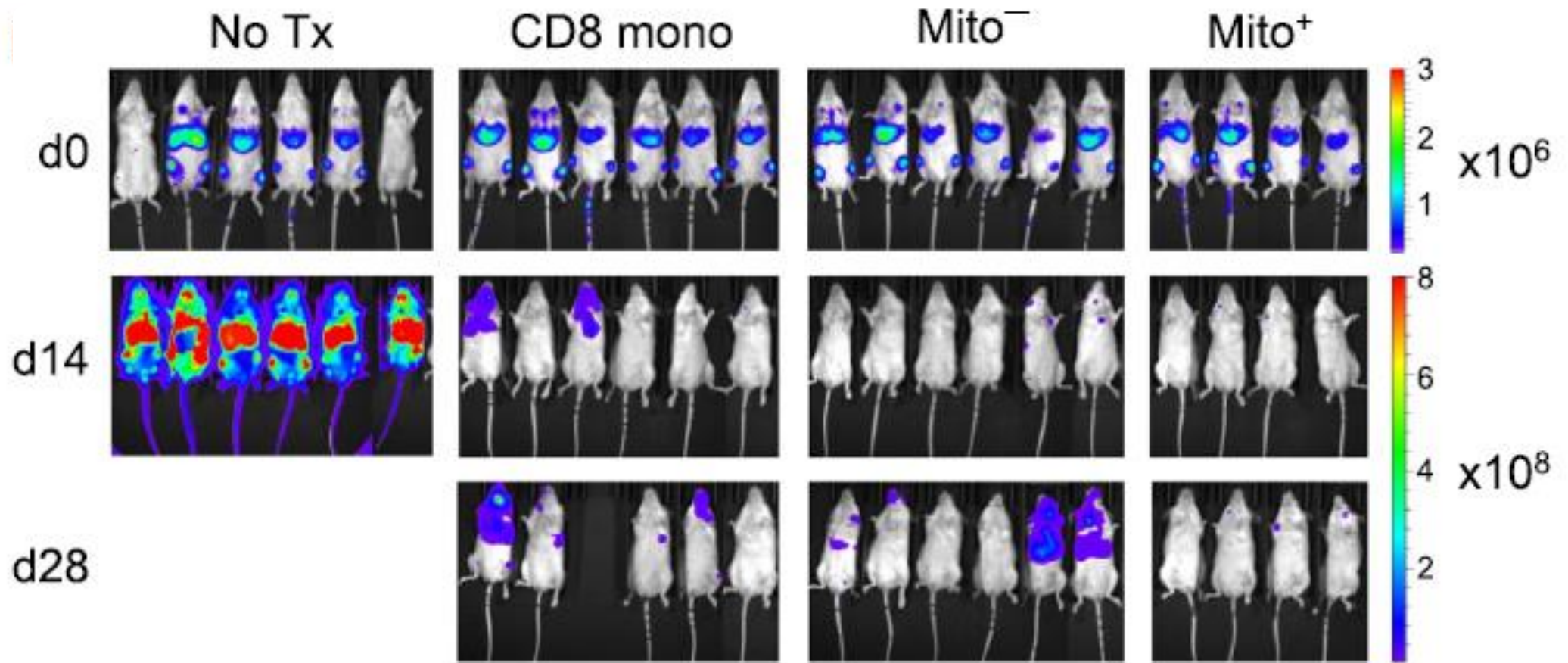
Correspondence

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shiladit@mit.edu (S.S.),
luca.gattinoni@lit.eu (L.G.)

In brief

Transfer of mitochondria via nanotubes from bone marrow stem cells to CD8⁺ T cells augments cellular metabolism, empowering engineered T cells and tumor-infiltrating lymphocytes to counteract exhaustion and fight tumors more effectively.

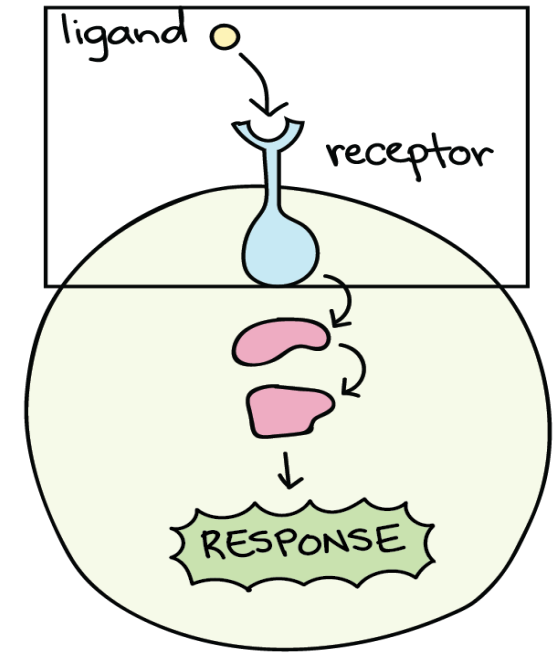
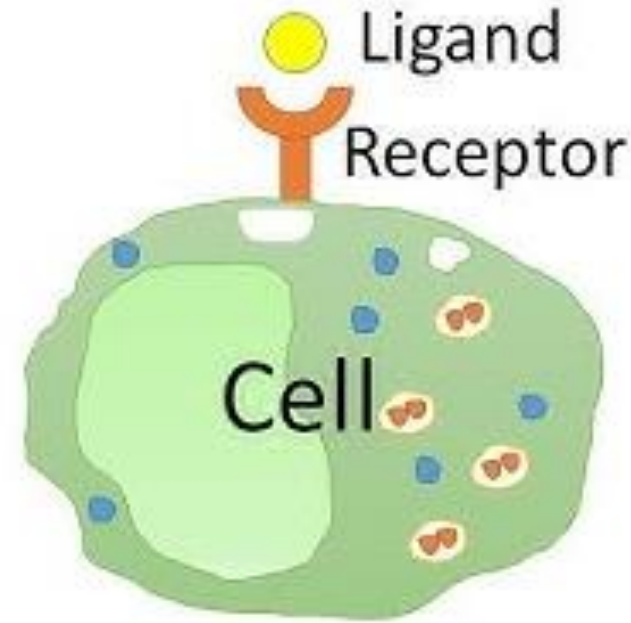
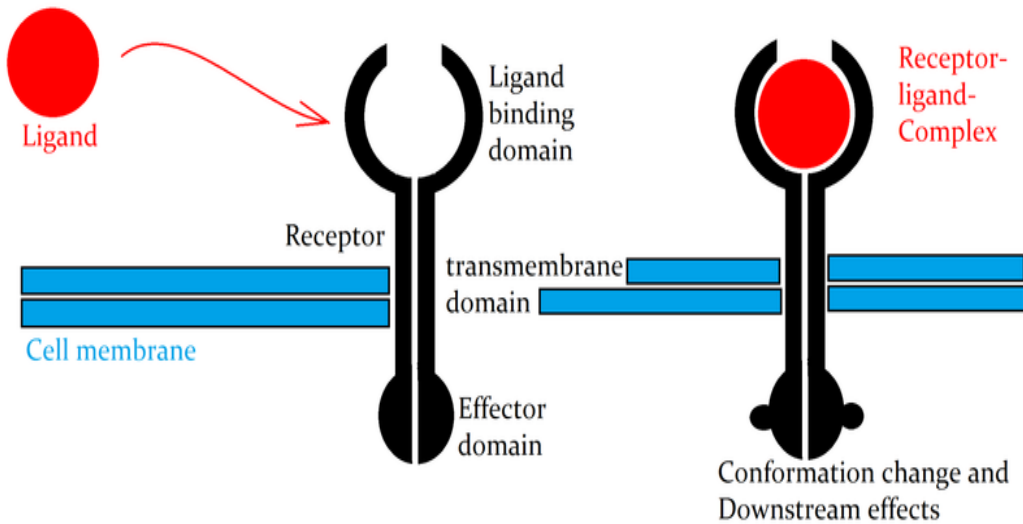
Mitochondrial transfer enhances human T cell antitumor immunity against leukemia



In vivo bioluminescent imaging and survival of NALM6-GL-bearing NXG mice (n = 4 or 6 mice/group).

No Tx, no treatment (n = 6)

Ligand and Receptors:



- **Ligand** is a substance that forms a **complex** with a biomolecules to serve a biological purpose *delivering a signal* in the process. Ligands can thus be thought of as *signaling molecules*.
- Ligand binding to a receptor protein alters the conformation by affecting the three-dimensional shape orientation.
- Ligands interact with proteins in target cells, which initiates some downstream *chemical signals*; these proteins are called **receptors**.